nature aging

Article

https://doi.org/10.1038/s43587-026-01225-9

Exercise enhances aged chondrocyte health through microRNA-29-enriched extracellular vesicles

Received: 20 May 2023

Accepted: 20 August 2026

Published online: 29 September 2026

Check for updates

Hirotaka Iijima[1,2,3] ✉, Kandarp M. Dave[1,2,3], Kai Wang[1,2,3], Adam M. Khay[1,3], Ella D'Amico[4], Wan-Yee Tang[5], Renee J. Rogers[6], John M. Jakicic[6] & Fabrisia Ambrosio[1,2,3] ✉

Extracellular vesicles (EVs) have been proposed to be mediators of the health-promoting effects of exercise throughout the body, but how EVs communicate these beneficial signals to recipient cells remains unknown. Here we interrogated these mechanisms using articular cartilage as a model. Network propagation of EV microRNA perturbations on a cartilage-specific network identified microRNA-29 as an exercise-responsive regulator of cellular aging. Consistent with this prediction, a 3-month aerobic exercise intervention in older adults increased microRNA-29 levels in EVs relative to baseline EVs. In vitro studies revealed that these exercise-primed EVs epigenetically de-repressed the *KL* gene, which encodes the longevity-associated protein α-Klotho, and restored a more youthful chondrocyte phenotype. EVs engineered to express microRNA-29 recapitulated these anabolic effects in a *KL*-dependent manner. Intra-articular administration of exercise-primed EVs into aged mice enhanced cartilage health, thereby confirming physiological and potentially translational relevance. These findings elucidate mechanisms by which exercise transduces beneficial signals in EVs to promote health in distal tissues through regulation of *KL*.

Extracellular vesicles (EVs) have emerged as a potent mediator through which the beneficial effects of physical exercise are transmitted throughout the body[1]. EVs are nanovesicles that contain functional biomolecular cargo, such as lipids, proteins and nucleic acids[2], which respond dynamically to physiological stimuli and environmental stressors[3,4]. Exercise-induced alterations in circulating EV cargo are increasingly being recognized as a potential mechanism underlying the systemic benefits of physical activity. Indeed, several studies to date have shown that aerobic exercise modulates the bioactivity of circulating EVs, rendering EVs therapeutically beneficial in the context of diseases such as cardiac ischemia–reperfusion injury, obesity and type 2 diabetes mellitus[5,6]. We aimed to better understand the molecular mechanisms through which exercise-responsive EVs regulate tissue homeostasis and contribute to the health-promoting effects of exercise.

In silico computational approaches provide a powerful framework for inferring the biological impact of EV cargo on recipient cells. This is

[1]Discovery Center for Musculoskeletal Recovery, Schoen Adams Research Institute at Spaulding, Charlestown, MA, USA. [2]Department of Physical Medicine & Rehabilitation, Harvard Medical School, Boston, MA, USA. [3]Department of Physical Medicine & Rehabilitation, Spaulding Rehabilitation Hospital, Charlestown, MA, USA. [4]Department of Rehabilitation and Regenerative Medicine, Columbia University Irving Medical Center, New York, NY, USA. [5]Department of Environmental and Occupational Health, University of Pittsburgh, Pittsburgh, PA, USA. [6]Department of Internal Medicine, Division of Physical Activity and Weight Management, University of Kansas Medical Center, Kansas City, KS, USA. ✉e-mail: hiijima1@mgh.harvard.edu; fambrosio@mgh.harvard.edu

particularly true for microRNAs (miRNAs) carried by EVs, because these regulate post-transcriptional gene expression through complementary binding to target mRNAs[7]. As such, the miRNA regulatory network has been used as a tool for dissecting the complex regulation by EVs on target biological processes[8,9]. Previous miRNA regulatory network analyses, however, have primarily focused on direct miRNA–target interactions and have not fully captured the downstream consequences of miRNA-mediated perturbations across global gene regulatory networks. To simulate the downstream effects of EV miRNAs, network propagation is a valuable tool for elucidating the global transcriptional responses they induce. This approach is further enhanced by incorporating tissue specificity, given that the precise actions of genes often depend on the tissue in which the cells reside[10]. Indeed, human diseases result from the disordered interplay of tissue-specific processes[11]. Tissue-specific network models clarify biological functions of individual genes to help identify disease driver genes that could exert important functional roles in the tissue of interest.

With this in mind, this study employed network medicine paradigms to simulate perturbations in miRNA regulatory networks induced by exercise-responsive EV cargo and identify molecular mechanisms underlying the beneficial effects of exercise. This study leveraged a cartilage-specific gene regulatory network, enabling precise modeling of how circulating EVs impact the cellular aging process in the cartilage microenvironment. Because age-related disruption of cartilage homeostasis is a key driver of knee osteoarthritis (KOA)[12], one of the leading causes of disability worldwide[13], cartilage serves as a valuable model for investigating the molecular mechanisms through which exercise-responsive EV cargo promotes tissue health. Given that exercise is recommended as a first-line intervention for the prevention and management of KOA[14], elucidating these mechanisms might not only advance our understanding of exercise-mediated cartilage protection[15], but also inform the development of exercise-primed EVs as regenerative rehabilitation therapeutics for musculoskeletal disease[16].

Results

Aerobic exercise upregulates miRNAs that modify chondrocyte–matrix adhesion processes

To date, most studies assessing the effects of exercise on EV biomolecules have focused on isolated miRNAs of interest, thereby precluding a holistic understanding of exercise-sensitive EV miRNAs and their downstream effects on recipient cells. To address this issue, we first performed a systematic literature search (Fig. 1a) and summarized EV miRNAs that are differentially regulated by exercise in older adults, as quantified by small RNA sequencing (small RNA-seq). We identified only one study that performed small RNA-seq on circulating EVs in our target population, older adults, before and after exercise[3]. In that study, male participants ($n = 10$; mean age, 69.3 years) completed an acute bout of aerobic exercise, and circulating EVs were collected before, immediately after and 3 h after exercise[3] (Fig. 1a). Participants were separated into two groups on the basis of baseline physical activity level (sedentary versus trained; $n = 5$ per group). The authors demonstrated that aerobic exercise primarily changed levels of miRNAs targeting the IGF-1 signaling pathway[3]. To assess the global EV miRNA response to acute aerobic exercise, we accessed and evaluated the raw small RNA-seq data (GSE144627) including all ten male participants in the study. Unlike the original study, which analyzed the sedentary and active groups separately[3], we combined both groups in our analysis to maximize sample size. After low-count filtering, 445 miRNAs were retained, of which 16 were differentially expressed after exercise ($|\log_2$ fold change$| > 1.0$, $P < 0.05$; Fig. 1b), whereas the original study identified nine and seven miRNAs that were differentially expressed in the sedentary and trained groups, respectively[3].

Next, we sought to determine the primary target genes of the 16 miRNAs. For this purpose, we used two databases, miEAA.2.0 (ref. 17) and miRTarBase[18], for target gene prediction (Fig. 1c). The analyses identified 43 genes implicated in the process of cell–matrix adhesion and extracellular matrix (ECM) remodeling, including *ITGB1* and *COL4A1* (ref. 19), both of which are targets of the miR-29 family[20]. Of note, when miR-4433b-3p, the only miRNA downregulated by exercise, was used as an input, no target gene was consistently predicted. As such, all the identified 43 target genes were the products of the remaining 15 miRNAs that were upregulated after aerobic exercise.

We then evaluated the biological function of these primary target genes. Given our focus on osteoarthritic cartilage, we generated a cartilage-specific functional gene network from the 43 target genes using HumanBase, which integrates large-scale genomic datasets to predict tissue- and cell-type-specific gene functions and functional interactions beyond existing literature biases[21]. Notably, the target genes formed a highly interconnected network (Fig. 1d), with *IL6* acting as a local hub. Genes such as *ITGB1* and *COL4A1* were also included, indicating that exercise-sensitive miRNA targets converge on coordinated biological processes in the cartilage system.

Functional enrichment analysis of the cartilage-specific network revealed Gene Ontology (GO) terms predominantly related to cell–matrix adhesion processes, including 'cell–substrate adhesion' and 'cell–matrix adhesion' (Fig. 1e). In articular cartilage, altered cell–matrix interactions initiate mechanotransductive processes, activating downstream signals that direct the transcription of genes involved in the regulation of cellular and tissue homeostasis[22]. Together, these in silico analyses revealed that the miRNAs that are significantly upregulated after aerobic exercise primarily affect chondrocyte–matrix adhesion and initiate downstream signaling events.

Perturbation of the cartilage-specific miRNA regulatory network implicates EVs in the regulation of cartilage aging

To infer the downstream effects of the upregulated EV miRNAs on the signaling cascade in articular cartilage (that is, a downstream effect of chondrocyte–matrix adhesion), we implemented a network-propagation approach to the cartilage-specific global functional gene network (Fig. 2a). The network paradigm introduced here assumes that there is an interaction between articular cartilage and circulating EVs that have penetrated into the joint cavity through the synovium, where intercellular gaps of the blood–joint barrier (0.1–5.5 µm) are greater than EVs (~0.15 µm)[23]. Systemically administered labeled EVs were detectable in the knee joint 48 h later (see inset in Fig. 2a), indicating that circulating EVs are able to reach the joint.

To simulate the EV–cartilage communication, we again constructed a cartilage-specific network using HumanBase[21]. With this global network in hand, we used a random-walk-restart (RWR) algorithm to identify the most frequently visited gene nodes on a random path in the cartilage-specific network[24]. The 43 target genes were used as seeded (starting) genes. This approach identified genes and pathways predicted to respond to miRNA perturbation (Fig. 2b). Figure 2c shows an example of in silico activation of *ITGB1*, a primary target of two exercise-upregulated miRNAs, miRNA-29a-3p and miRNA-29b-3p[20], on the network. Applying RWR to the cartilage-specific network revealed that perturbation of target genes of miRNAs carried by EVs led to pseudo-activation (affinity score > 0) of key cartilage homeostasis genes, such as *ACAN*, *DCN*, *GDF5* and *COL9A3*. Extending this analysis, RWR was applied repeatedly to each of the 43 EV miRNA-target genes on a cartilage-specific network, revealing that the pseudo-activated genes were highly associated with 'integrin signaling', 'focal adhesion', 'TGF-β signaling', 'PI3K–Akt signaling' and 'ECM–receptor interaction' (Fig. 2d), pathways previously implicated in the pathogenesis of KOA[25]. Supporting this, disease-enrichment analysis of the pseudo-activated genes demonstrated a strong association with KOA and idiopathic, age-related osteoarthritis, whereas post-traumatic osteoarthritis showed comparatively lower enrichment and impact (Extended Data Fig. 1). These findings indicate that exercise-primed EV miRNAs regulate biological processes of age-related KOA.

https://doi.org/10.1038/s43587-026-01225-9

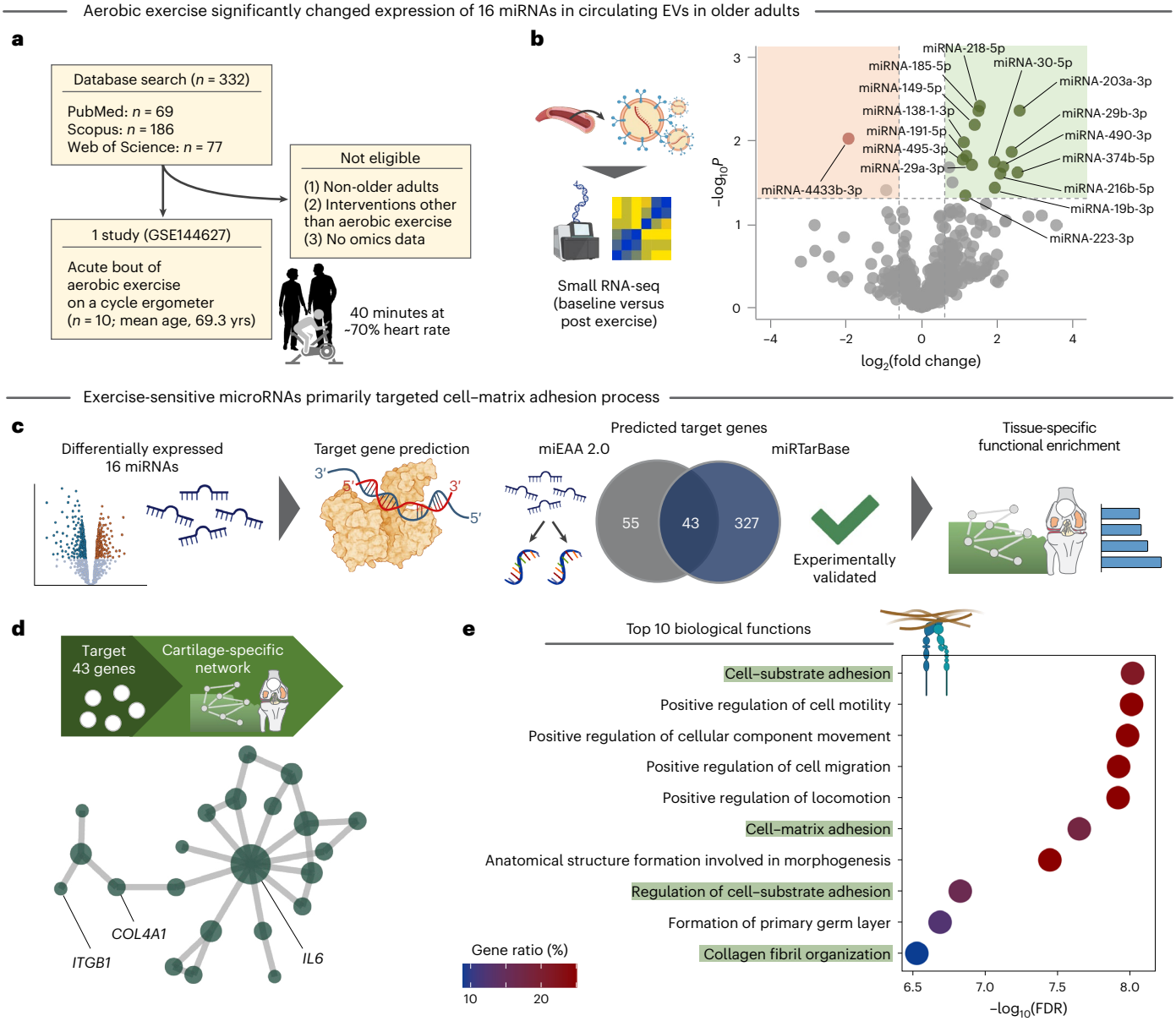


Fig. 1 | Small RNA-seq revealed that an acute bout of aerobic exercise significantly changed 16 EV miRNAs that primarily targeted the cell–matrix adhesion process. a, An electronic database search yielded a total of 332 studies, of which 1 study (GSE144627) was included[3]. The included study performed small RNA-seq from circulating EVs before and after an acute bout of aerobic exercise in older adults ($n = 10$; mean age, 69.3 years)[3]. **b**, Small RNA-seq identified 16 differentially expressed miRNAs ($-\log_{10}$(unadjusted nominal two-sided P value) > 1.3 and $|\log_2$ fold change$|$ > 1.0), 15 of which were upregulated following exercise. **c**, Target gene prediction using two databases (miEAA 2.0 and miRTarBase) identified 43 robust target genes of the 16 exercise-responsive miRNAs. **d**, Cartilage-specific functional gene network, constructed using HumanBase[21], demonstrating that the 43 target genes, including *IL6*, *ITGB1* and *COL4A1*, form a highly interconnected network, indicative of coordinated biological processes within the cartilage system. **e**, The cell–substrate adhesion process was the primary biological function significantly (false discovery rate (FDR)< 0.05) associated with the 43 target genes, as indicated by GO enrichment analysis performed on the cartilage-specific functional network. Only the top 10 GOs were provided for illustration purposes. The green highlighted pathways represent processes related to cell–matrix interactions. Statistical analysis was performed using the R package DESeq2 based on a two-sided Wald test (**b**). All raw data supporting these findings are available in the Supplementary Data. Data visualizations created in BioRender; Iijima, H. https://BioRender.com/f55bqvo (2026).

Studies have shown that aging alters ECM biophysical properties, with increased stiffness emerging as a hallmark of aged tissues[26,27]. This is particularly evident in articular cartilage, in which aged cartilage exhibits substantially greater stiffness than does its young counterpart (Young's modulus (E) = ~50–100 kPa in aged cartilage versus E = ~1–30 kPa in young)[28]. Increased ECM stiffness promotes aberrant ECM remodeling and disrupts chondrocyte function through mechanotransductive pathways[25,29–32], contributing to KOA. Recent findings further demonstrated that an aged-like stiff ECM induces an aged chondrocyte phenotype through adhesion-mediated mechanotransduction and epigenetic repression of the longevity-associated gene *KL* (or *KLOTHO*), accompanied by morphological changes linked to focal adhesion kinase[25,33]. Building from this previous work, our findings from in silico analyses led us to ask whether exercise-primed EVs counteract the morphological and functional changes in chondrocytes induced by an aged-like (stiff) ECM. The experiments described in the following sections were designed to address this question.

https://doi.org/10.1038/s43587-026-01225-9

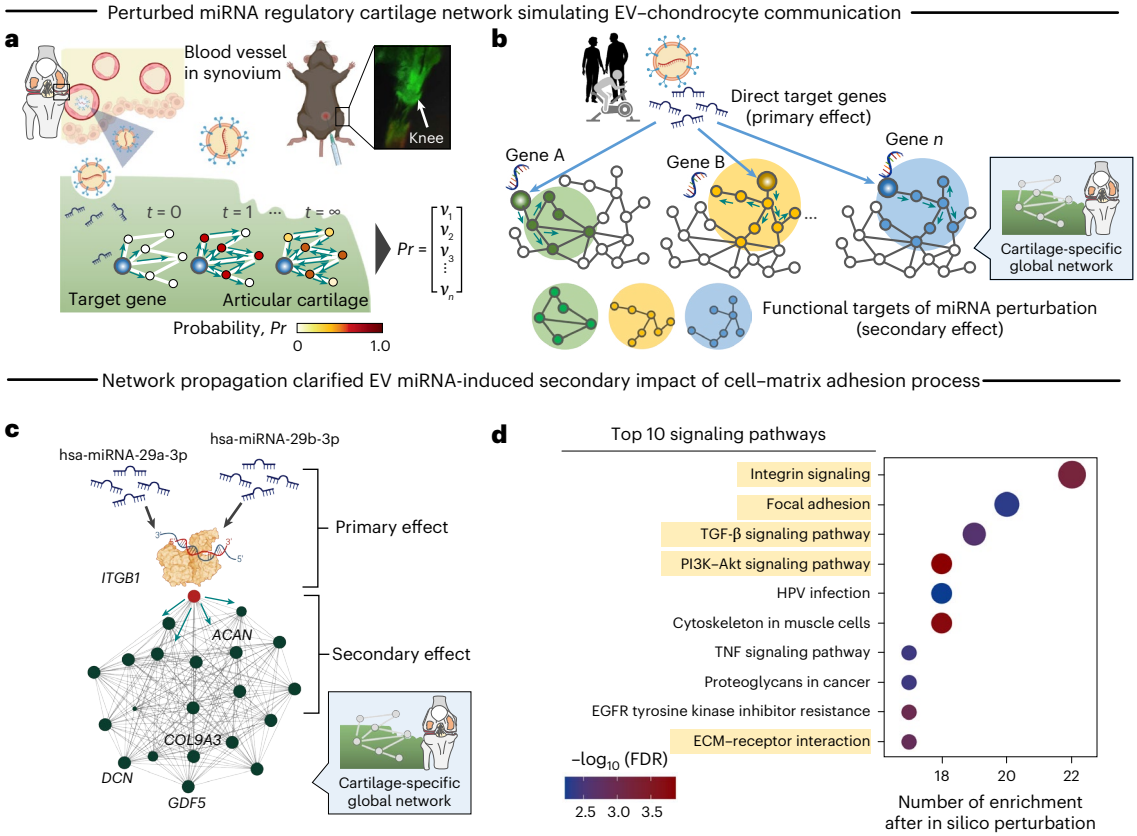


Fig. 2 | In silico perturbation of the miRNA regulatory network revealed that exercise-primed EV miRNAs drive downstream signaling, with dominant effects on cartilage aging pathways. a, Schematic of the in silico perturbation model simulating EV–cartilage communication in the knee joint. The model assumes circulating EVs penetrate the joint cavity through the synovium, validated by localized fluorescence of systemically injected labeled EVs in murine knees (inset). Network propagation was initiated at each EV miRNA target gene to infer downstream impacts on the cartilage-specific functional network. V_n represents the probability score assigned to gene n. **b**, Schematic of the iterative network propagation strategy within a cartilage-specific global network. Each direct (primary) target gene of EV miRNAs (for example, Gene A, Gene B and Gene n) was used individually as a seed node, and network propagation was performed independently to model the spread of influence to neighboring genes. This generated a perturbed subnetwork representing inferred secondary (functional) effects. The procedure was repeated for all primary targets, and the resulting subnetworks were integrated to define the overall downstream impact of EV miRNAs, on the basis of the principle that functionally related genes cluster in biological networks. **c**, Example of a perturbed miRNA regulatory network of *ITGB1*, a primary target gene of EV miRNAs (hsa-miRNA-29a-3p, hsa-miRNA-29b-3p). To infer phenotypic targets, 50 vicinity genes (the 50 highest probability scores) were used for the subsequent annotation of signaling pathways. **d**, Perturbation of miRNA regulatory network identified specific Kyoto Encyclopedia of Genes and Genomes (KEGG) signaling pathways as a phenotypic target of exercise-primed EV miRNAs. The yellow highlighted pathways represent signaling previously implicated in the pathogenesis of age-related KOA[25]. All raw data supporting these findings are available in the Supplementary Data. Data visualizations created in BioRender; Iijima, H. http://BioRender.com/f55bqvo (2026).

Exercise-primed EVs override chondrocyte morphological alterations induced by a stiff ECM

We first conducted an exercise study in older individuals. Circulating EVs were isolated from older adults aged 65–85 years (mean, 69.2 ± 3.7 years; $n = 10$; 80% female participants) who completed 3 months of an aerobic exercise protocol that aimed to achieve moderate intensity on the basis of perceived exertion (Fig. 3a). Body weight remained stable over the 3-month intervention, with mean body weight ± s.d. of 83.2 ± 15.9 kg at baseline and 83.4 ± 16.6 kg at 3 months (mean difference, 0.2 ± 2.4 kg; $P = 0.830$), consistent with findings from previous studies involving moderate-intensity aerobic exercise interventions[34].

EVs were isolated using size-exclusion chromatography before and after 12 weeks of exercise (EV^{pre} and EV^{post}, respectively) using a previously described protocol and following MISEV guidelines[35,36]. As determined by NanoSight nanoparticle tracking analysis, the average size and number of EVs were similar between groups (Fig. 3b,c), consistent with findings from previous exercise cohorts[37]. EV purity was confirmed by western blotting for established EV markers (CD63 and CD9), low albumin contamination, and absence of calnexin (Fig. 3d and Extended Data Fig. 2). Together, these findings support the successful isolation of circulating EVs with acceptable purity.

Next, EV^{pre} or EV^{post} were cultured with chondrocytes from an aged male donor (65 years old; body mass index (BMI), 23.4 kg m^{-2}) seeded onto a substrate engineered to mimic the matrix stiffness of aged human cartilage (100 kPa)[28]. We used this experimental design to test the ability of the exercise-primed EVs to counteract the age-related pathogenic phenotype induced by an aged-like stiff substrate. Before analysis, we confirmed the rapid uptake of fluorescently labeled EVs by aged human chondrocytes (Fig. 3e). No significant difference in internalization was observed between EV^{pre} and EV^{post} ($P = 0.192$) (Fig. 3e; blue dots). We also assessed EV uptake on a young-like soft substrate to evaluate the potential influence of matrix stiffness on EV internalization, and we found similar results under this condition ($P = 0.317$) (Fig. 3e; yellow dots).

To investigate the ability of exercise-primed EVs to modulate chondrocyte–matrix adhesion processes, we assessed morphological alterations of chondrocytes cultured on a stiff substrate that were exposed to either EV^{pre} or EV^{post} (Fig. 3f). Specifically, we quantified

chondrocyte morphology across 53 morphological variables using CellProfiler[38] (Supplementary Table 1). Figure 3f provides representative images of chondrocyte morphology, assessed through phalloidin staining across the different groups. We posited that exercise-primed EVs would interfere with the aged phenotype induced by a stiff matrix and drive a more youthful chondrocyte morphological phenotype that resembles chondrocytes seeded on a soft substrate[25]. As such, the features of chondrocytes cultured on a soft substrate were used as a reference for a youthful profile.

Principal component analysis (PCA) confirmed clear differences between the morphology of chondrocytes treated with EV^{pre} compared with that of chondrocytes treated with EV^{post} (Fig. 3g). Notably, the morphology of chondrocytes seeded on a stiff matrix in the presence of EV^{post} approximated the morphology of cells cultured on a soft substrate (Fig. 3g). Figure 3h summarizes the changes in individual morphological features. Of the 19 morphometric variables clearly altered by EV^{post}, the majority (12 (63.2%)) overlapped with the effect of a soft substrate. Among the individual morphological variables of interest, we found that EV^{post} increased chondrocyte 'FormFactor' (Fig. 3i). 'FormFactor' is a metric of cellular sphericity or roundness and was the feature previously identified in chondrocytes as being highly sensitive to mechanical input[25]. However, aged chondrocytes treated with EV^{pre} did not display clear morphometric changes and more closely resembled untreated cells cultured on a stiff substrate (Fig. 3f–i).

Given that integrin-mediated cell–matrix adhesion represents a central mechanotransductive pathway activated by a stiff ECM[25], we next examined whether exercise-primed EVs modulate integrin signaling in aged human chondrocytes. Among integrin family members, ITGB1 functions as a hub subunit that pairs with multiple α integrins and mediates chondrocyte adhesion to collagen-rich ECM and downstream mechanotransductive signaling[39–41]. Accordingly, chondrocytes cultured on a stiff substrate upregulated the target genes of exercise-sensitive microRNAs, among which *ITGB1* was identified as a top-ranking gene (Extended Data Fig. 3a,b). In line with the observed morphological rescue (Fig. 3f–i), ITGB1 expression was significantly reduced in aged chondrocytes cultured on a stiff substrate with EV^{post} compared with that in EV^{pre}-treated chondrocytes (Fig. 3j). This finding is particularly notable given that ITGB1-dependent focal adhesion signaling functions upstream of focal adhesion kinase activation, a pathway that we previously identified as a driver of aberrant mechanotransduction in chondrocytes[25]. Together, these data suggest that exercise-primed EVs attenuate stiffness-induced chondrocyte–matrix adhesion signaling, at least in part through downregulation of ITGB1. This finding provides mechanistic support for our network-based inference identifying chondrocyte–matrix adhesion as a primary target of EV-associated miRNAs (Fig. 1c–e).

Exercise-primed EVs counteract compromised chondrocyte integrity and *KLOTHO* promoter hypermethylation induced by a stiff ECM

To evaluate the ability of exercise-primed EVs to promote chondrocyte health, we repeated the above in vitro experiment, this time testing whether exercise-primed EVs restore a more youthful phenotype to aged chondrocytes. We found that aged chondrocytes treated with EV^{post} exhibited increased expression of chondrogenic markers (type II collagen, aggrecan and SOX9) and reduced expression of catabolic markers (MMP13 and ADAMTS4) compared with EV^{pre}-treated chondrocytes. EV^{post} shifted chondrocytes toward a soft (youthful-like) phenotype, whereas EV^{pre}-treated chondrocytes remained similar to untreated controls (Fig. 4a). When these five anabolic and catabolic outcome measures were mapped onto a PCA plot, the pattern was consistent with the morphometric observations: EV^{post}-treated chondrocytes closely resembled cells on a soft substrate, whereas EV^{pre}-treated cells approximated stiff substrate conditions (Fig. 4b). This finding indicates that exercise-primed EVs can shift the anabolic–catabolic balance toward an anabolic state, thereby promoting a youthful chondrocyte phenotype (Fig. 4c).

We then explored candidates that may drive the beneficial effect of EV^{post} on chondrocyte integrity. Our recent work demonstrated that a stiff substrate increases methylation of the *KLOTHO* promoter in chondrocytes[25]. *KLOTHO* encodes α-Klotho, a longevity-associated protein and upstream regulator of PI3K–Akt signaling[42] that attenuates age-related decline in articular chondrocytes[25]. In response to a stiff substrate, transcriptional machinery is recruited to the *KLOTHO* promoter in murine chondrocytes, inhibiting the downstream geroprotective cascade[25]. One such component of transcriptional machinery is DNMT1, an enzyme that catalyzes the transfer of methyl groups to CpG dinucleotides, usually repressing gene transcription by altering chromatin structure and blocking the access of transcription factors at the gene regulatory region[43]. Consistent with this, reanalysis of archived transcriptomic data from primary murine chondrocytes cultured on young-like soft (0.5–1 kPa) versus aged-like stiff (100 kPa) substrates (GSE63615)[32] revealed that DNMT1 expression was significantly elevated under stiff conditions (Extended Data Fig. 3b).

In aged chondrocytes treated with EV^{post}, DNMT1 levels were significantly decreased compared with those in EV^{pre}-treated counterparts (Fig. 4d), whereas levels of α-Klotho were increased in EV^{post}-treated chondrocytes (Fig. 4d); accordingly, changes in DNMT1

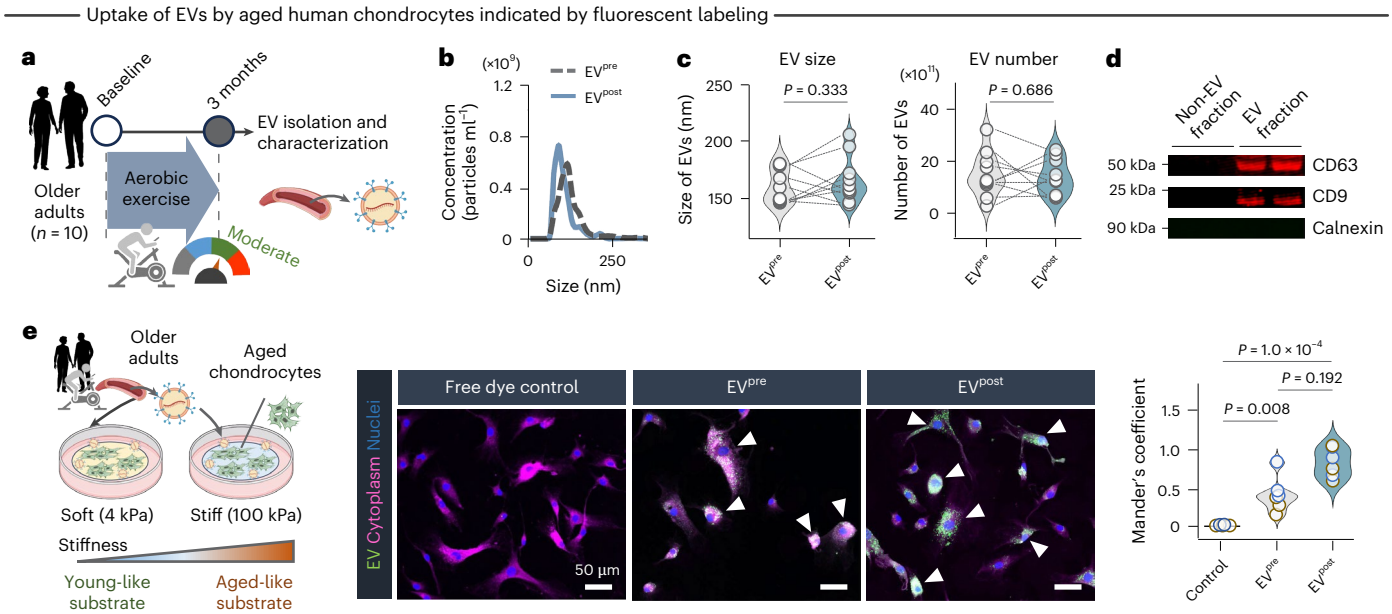
Fig. 3 | Exercise-primed EVs over-rode chondrocyte morphometric alterations induced by aged-like stiff substrate. a, Schematic showing the clinical-trial design. Circulating EVs were isolated from serum before and after a 3-month period of a moderate-intensity aerobic exercise protocol in older adults ($n = 10$). **b**, Representative nanoparticle tracking analysis (NTA) profiles of EV^{pre} and EV^{post}. **c**, Quantification of EV size and number assessed by NTA ($n = 10$ per group), showing comparable properties between EV^{pre} and EV^{post}. **d**, Western blot validation of EV purity. EV markers CD63 and CD9 were enriched in EV fractions, whereas intracellular marker calnexin was absent. Samples shown in each blot were derived from the same experiment and processed in parallel. **e**, Assessment of EV uptake in vitro. Primary human chondrocytes were cultured on substrates mimicking young-like soft (4 kPa) or aged-like stiff (100 kPa) tissue. Cells were then exposed to fluorescently labeled EV^{pre} or EV^{post}, and EV uptake was quantified using Mander's coefficient, a standard metric of colocalization[82]. Colocalization with MitoTracker, highlighted by white arrowheads, demonstrated greater intracellular uptake of EVs compared with free dye controls ($n = 6$ per group). Data in soft substrates are shown in yellow and stiff substrates in blue. **f**, Schematic of the CellProfiler-based workflow for quantitative assessment of chondrocyte morphology (53 morphometric variables) and representative phalloidin-stained images of chondrocytes cultured on soft or stiff substrates, with EV^{pre} or EV^{post} administered exclusively to cells cultured on the stiff substrate. **g**, PCA of chondrocyte morphology based on 53 CellProfiler-derived morphometric variables, showing a clear separation between EV^{pre}- and EV^{post}-treated chondrocytes and a shift of EV^{post}-treated cells toward the morphological profile of chondrocytes cultured on a young-like soft substrate ($n = 4$ per group). **h**, EV^{post} significantly altered 19 individual morphometric variables compared with EV^{pre}, with the majority of changes (12 variables) overlapping with those observed in chondrocytes cultured on a young-like soft substrate versus an aged-like stiff substrate. **i**, EV^{post} significantly increased chondrocyte FormFactor (that is, higher roundness) compared with EV^{pre} ($n = 4$ per group; 60–110 cells per individual sample). **j**, EV^{post} significantly reduced ITGB1 expression compared with EV^{pre} ($n = 10$ per group; 50–150 cells per individual sample). Statistical analysis was performed using a two-tailed paired *t*-test (**c**,**i**,**j**) or analysis of variance with post-hoc Tukey–Kramer test (**e**). For violin plots, values obtained from chondrocytes cultured on soft (gray dotted line) and stiff substrates (orange dotted line) are shown as reference lines. All raw data, including uncropped western blot images, are available in the Supplementary Data. Data visualizations created in BioRender; Iijima, H. http://BioRender.com/f55bqvo (2026).

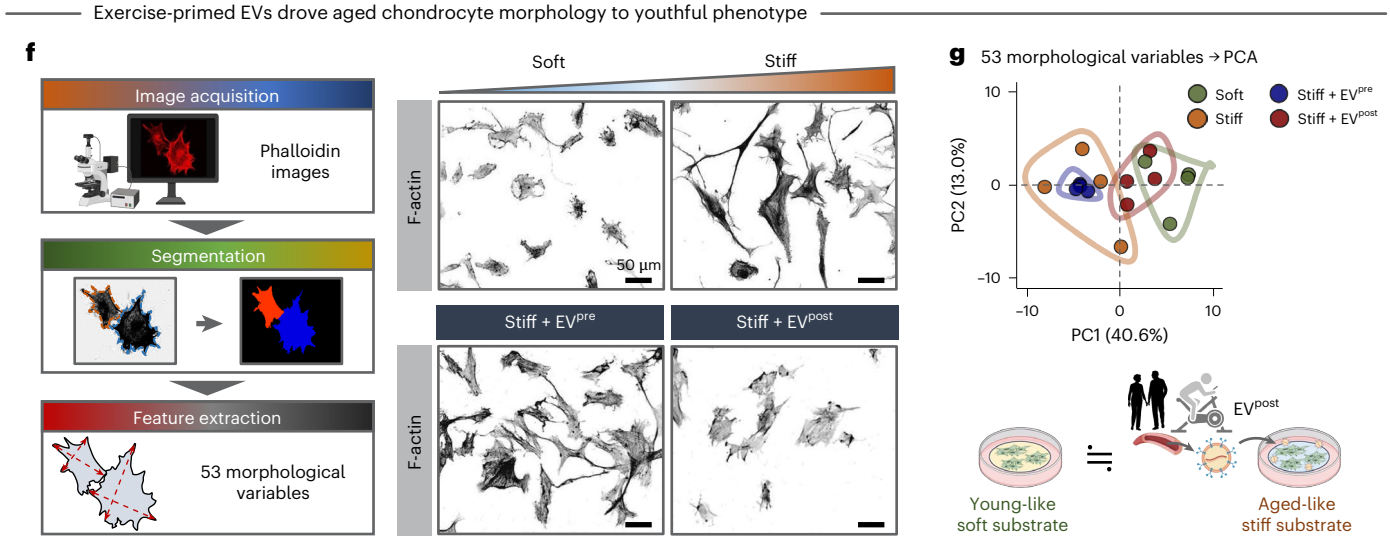
https://doi.org/10.1038/s43587-026-01225-9

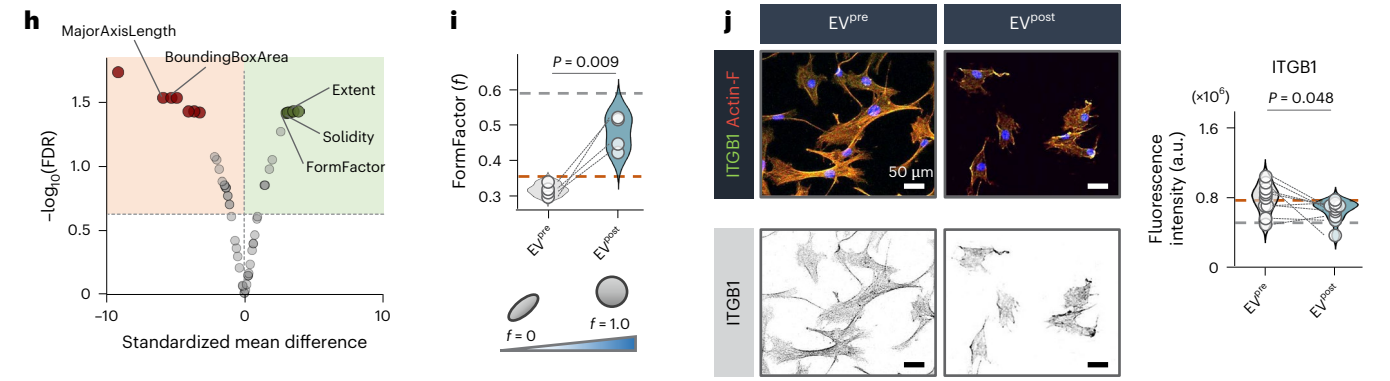
and α-Klotho (ΔPost–Pre) were significantly negatively correlated (Fig. 4e). These changes were further accompanied by upregulated expression of *KLOTHO* mRNA and decreased *KLOTHO* promoter methylation (Fig. 4f). EV^{post} treatment also led to increased global DNA demethylation (Fig. 4f). These cumulative findings suggest that aerobic-exercise-primed EVs can counteract aged chondrocyte hypermethylation, including *KLOTHO* promoter hypermethylation, induced by a stiff ECM.

Exercise-primed EVs mitigate age-related KOA in vivo

To evaluate the physiological relevance of our in vitro experiments, we next evaluated the effects of aerobic-exercise-primed EVs in a murine model of age-related KOA[25,44]. We first confirmed EV uptake by chondrocytes in articular cartilage of the aged murine knee joint ex vivo (Fig. 5a), supporting the capacity of circulating EVs to deliver miRNA cargo to chondrocytes. Next, aged male mice that were 21–22 months old, corresponding to approximately 56–69 years in humans[45],







https://doi.org/10.1038/s43587-026-01225-9

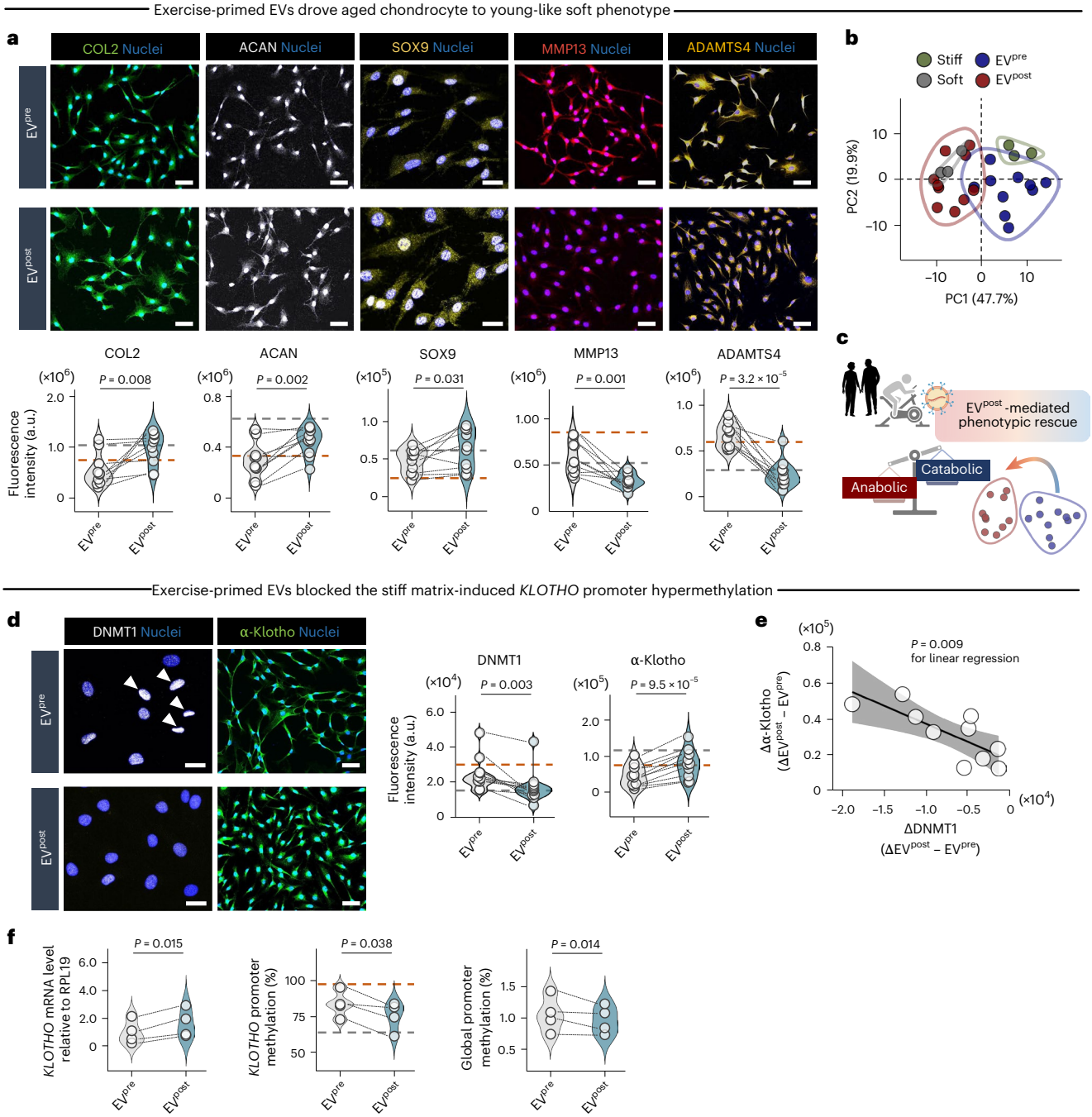


Fig. 4 | Exercise-primed EVs over-rode compromised chondrocyte integrity and *KLOTHO* promoter hypermethylation induced by aged-like stiff substrate. a, Representative immunofluorescence images and quantification showing that EV^{post} increased the expression of chondrogenic markers (type II collagen, aggrecan and SOX9) and reduced the expression of catabolic markers (MMP13 and ADAMTS4) in aged chondrocytes cultured on a stiff substrate, compared with EV^{pre} ($n = 10$ per group; 20–100 cells quantified per individual sample). Scale bars, 50 µm (all images except SOX9), 20 µm (SOX9 image). **b**, PCA integrating five molecular outcome measures (type II collagen, aggrecan, SOX9, MMP13 and ADAMTS4) demonstrating that EV^{post}-treated chondrocytes clustered closer to cells cultured on a soft (youthful-like) substrate, whereas EV^{pre}-treated chondrocytes approximated stiff substrate conditions. **c**, Schematic illustrating that exercise-primed EVs restore a youthful chondrocyte phenotype by rebalancing the anabolic–catabolic balance toward anabolic state. **d**, Immunofluorescence quantification showing that EV^{post} significantly reduced DNMT1 levels and increased α-Klotho expression in aged chondrocytes compared with EV^{pre} ($n = 10$ per group; 15–20 cells quantified per individual sample). Representative DNMT1-positive chondrocytes are indicated by white arrowheads. Scale bars, 50 µm (DNMT1), 20 µm (α-Klotho). **e**, Linear regression analysis of changes between chondrocytes treated with EV^{post} and EV^{pre} ($\Delta EV^{post} - EV^{pre}$), revealing a significant inverse relationship between DNMT1 and α-Klotho expression ($n = 10$). The black line indicates the fitted linear regression line, and the shaded gray area represents the 95% confidence interval of the estimated regression line. **f**, EV^{post} treatment increased *KLOTHO* mRNA expression and decreased *KLOTHO* promoter methylation in aged chondrocytes, accompanied by a more homogeneous pattern of global DNA demethylation ($n = 4$ per group). *KLOTHO* mRNA expression was first normalized to RPL19 and then standardized to EV^{pre} (mean expression in EV^{pre} set to 1) and is presented in arbitrary units. Statistical analyses were performed using a two-tailed paired *t*-test (**a**,**d**,**f**) and linear regression (**e**). For violin plots, values obtained from chondrocytes cultured on soft (gray dotted line) and stiff substrates (orange dotted line) are shown as reference lines. All raw data supporting these findings are available in the Supplementary Data. Data visualizations created in BioRender; Iijima, H. http://BioRender.com/f55bqvo (2026).

https://doi.org/10.1038/s43587-026-01225-9

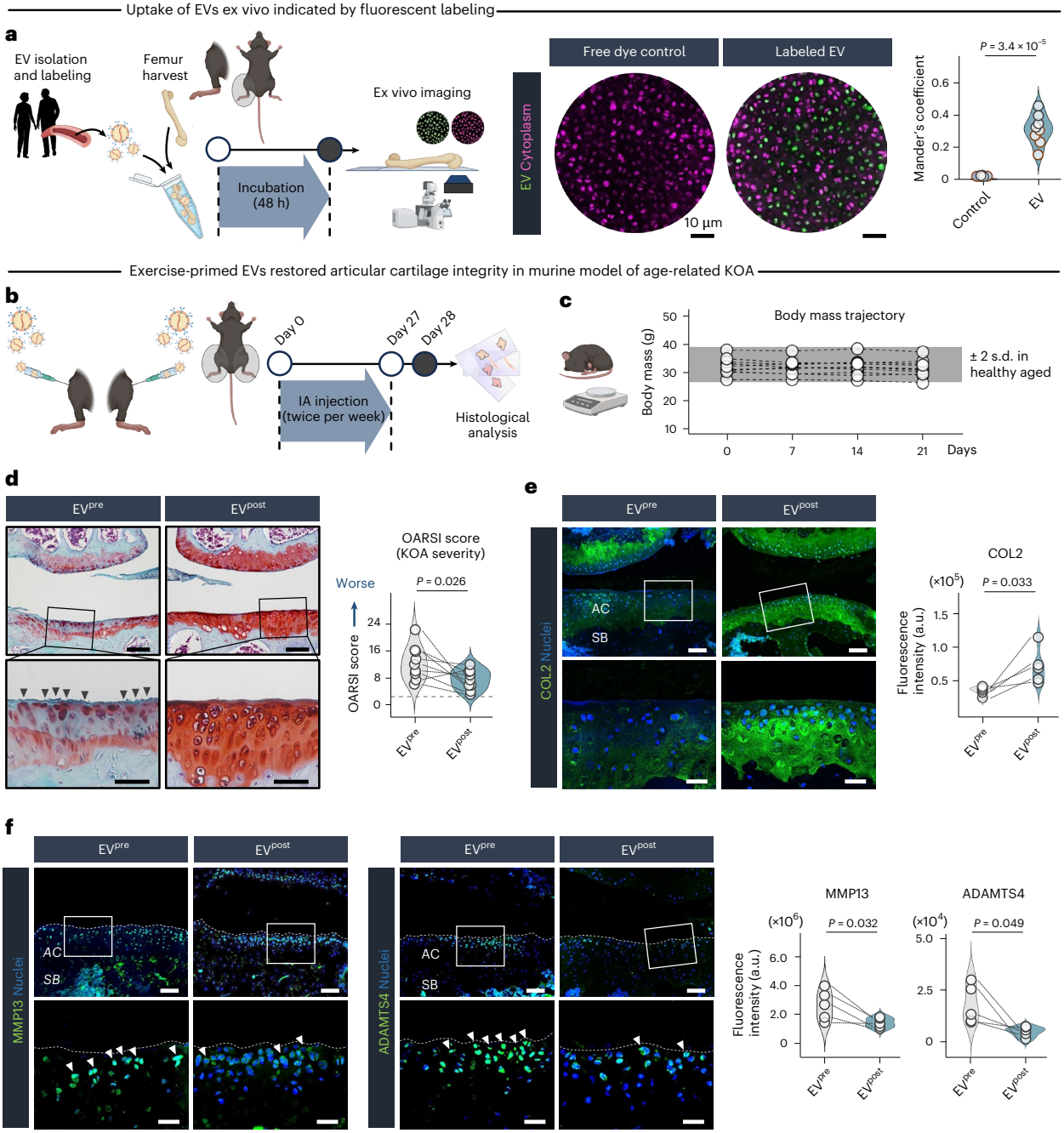


Fig. 5 | Exercise-primed EVs mitigated age-related cartilage degeneration in vivo. a, Ex vivo EV uptake assay. Fluorescently labeled circulating EVs were incubated with femoral articular cartilage collected from male young (black circle) and aged (orange circle) mice for 48 h, followed by confocal imaging to assess EV internalization by chondrocytes. Representative images and quantification using Manders' coefficient demonstrated significantly greater EV uptake in labeled EV-treated samples compared with free dye controls ($n = 4$ per group per age). **b**, Schematic of intra-articular EV administration in male aged mice. Male aged (21–22 months) mice received bilateral knee injections twice weekly for 4 weeks, with EV^{pre} injected into one knee and EV^{post} into the contralateral knee, followed by histological analysis. IA, intra-articular. **c**, Body mass of individual mice remained stable throughout the experimental period. The mean body-mass change was −0.62 ± 1.08 g, which falls within the expected distribution (±2 s.d.) for healthy aged mice (21–22 months old), represented by the gray shaded area. **d**, Representative Safranin O/Fast Green-stained knee joint sections and higher-magnification views showing reduced cartilage degeneration in EV^{post}-treated joints compared with their EV^{pre}-treated counterparts. Gray arrowheads indicate representative regions of degenerated cartilage. Quantification of Osteoarthritis Research Society International (OARSI) scores demonstrates significantly lower cartilage damage in EV^{post} knees within the same animals ($n = 9$ per group). Scale bars, 100 µm (top), 50 µm (bottom). **e**,**f**, Immunohistochemical analysis of cartilage anabolic (**e**) and catabolic (**f**) markers in aged mouse knees following EV treatment. Representative images and quantification demonstrate increased type II collagen expression together with reduced expression of the catabolic enzymes MMP13 and ADAMTS4 in EV^{post}-treated joints compared with EV^{pre}-treated joints ($n = 5$ per group). Representative MMP13- and ADAMTS4-positive chondrocytes are indicated by white arrowheads. Scale bars, 50 µm (top), 20 µm (bottom). AC, articular cartilage; SB, subchondral bone. The boxed regions in **d**–**f** are enlarged views of the corresponding areas. Statistical analyses were performed using a two-tailed Student's *t*-test (**a**) and a two-tailed paired *t*-test (**d**–**f**). All raw data supporting these findings are available in the Supplementary Data. Data visualizations created in BioRender; Iijima, H. http://BioRender.com/f55bqvo (2026).

received intra-articular injections of EV^{pre} or EV^{post} twice per week for 4 weeks, using our established lateral injection approach[46] (Fig. 5b). The approach was designed to mimic standard clinical procedures. To mitigate the effects of between-animal variability, we used a within-subjects design. In the experimental group, each mouse received injections of EV^{pre} into the left knee and injections of EV^{post} into the right knee. Body mass remained largely stable throughout the experimental period, with a mean change of −0.62 ± 1.08 g, which falls in the normal distribution (±2 s.d.) of healthy aged mice (Fig. 5c). These findings suggest that intra-articular EV administration did not induce systemic physiological stress or weight loss.

Compared with the contralateral knees that received EV^{pre}, those that received EV^{post} exhibited significantly lower cartilage degeneration, as determined through Osteoarthritis Research Society International scoring (Fig. 5d). Immunohistochemical analyses of markers of cartilage anabolic and catabolic activity demonstrated increased expression of type II collagen, together with reduced expression of the catabolic enzymes Mmp13 and Adamts4 in articular cartilage treated with EV^{post} compared with its EV^{pre}-treated counterpart (Fig. 5e,f). Consistent with in vitro observations, these results demonstrate that exercise-primed EVs can counteract the deleterious effects of a stiff, aged cartilage microenvironment in vivo, as determined by decreased cartilage degeneration, upregulation of anabolic markers and downregulation of catabolic markers in chondrocytes.

Engineered EVs overexpressing the miRNA-29 family recapitulate the chondrocyte health-promoting effects of exercise-primed EVs

We next sought to identify miRNAs capable of recapitulating the chondrocyte-health-promoting effects of exercise-primed EVs. Among the 16 identified exercise-responsive miRNAs, the miRNA-29 family emerged as a particularly compelling candidate because it is a 'master fibromiRNA regulator', targeting at least 16 genes related to the ECM, including many collagen isoforms and *ITGB1* (ref. 20). This function is unique to miRNA-29, given that no other miRNA targets more than 11 collagen genes[47]. In accordance with this distinctive regulatory capacity, the miRNA-29 family has demonstrated anti-fibrotic actions in varied animal models of tissue fibrosis[48–50], including chemically induced synovial fibrosis in murine knees[51].

To determine whether miRNA-29 functionally drives the regulatory activity of exercise-responsive EV miRNAs, we performed in silico deletion analyses. Deletion of *miRNA-29a-3p* and *miRNA-29b-3p* reduced the number of predicted EV miRNA-target genes from 43 to 7 (Fig. 6a), indicating that miRNA-29 accounts for most of the regulatory capacity of exercise-sensitive EV miRNAs. Circulating levels of miRNA-29 were significantly lower in individuals with KOA than in their healthy counterparts (Fig. 6b), supporting a clinical association between miRNA-29 deficiency and compromised cartilage integrity.

Next, to quantify the functional contribution of miRNA-29 to downstream signaling in cartilage, we again performed network-perturbation analysis following in silico miRNA-29 deletion, using the same cartilage-specific network-propagation framework described above for perturbation of EV miRNA-target genes (Fig. 6c). By calculating pathway-level differences between the full miRNA perturbation and the miRNA-29-deleted models, we estimated the specific impact of miRNA-29 on downstream signaling cascades (Fig. 6d). More than 75% of the full miRNA perturbation effects on key age-related KOA pathways, including 'integrin signaling', 'PI3K–Akt signaling', 'focal adhesion', 'TGF-β signaling' and 'ECM–receptor interaction,' were attributable to miRNA-29a-3p and miRNA-29b-3p. Notably, in our participants, both miRNA-29a-3p and miRNA-29b-3p in EVs were significantly upregulated following exercise (Fig. 6e). The upregulation of miRNA-29a-3p and miRNA-29b-3p after aerobic exercise is in line with the general understanding that exercise elicits anti-fibrotic effects[52]. Moreover, exercise-induced increases in these two miRNAs positively correlated with α-Klotho expression in chondrocytes (Fig. 6f). Inter-individual variability in miRNA-29 responses suggests that not only may EV miRNA-29 family members act as a key molecular mediator of exercise-induced cartilage protection, but they could also serve as a potential biomarker to distinguish responders from non-responders on the basis of their capacity to engage cartilage-protective signaling pathways (Fig. 6f).

To validate the findings from in silico analyses, we developed engineered EVs enriched with miRNA-29a-3p and miRNA-29b-3p and evaluated their functional impact on aged human chondrocytes in vitro. First, we confirmed efficient internalization of EV miRNAs, demonstrating robust uptake of labeled EV cargo in recipient chondrocytes (Fig. 7a). Given that integrin signaling emerged as one of the dominant miRNA-29-regulated pathways in our in silico perturbation analysis, we next examined the effect of miRNA-29 on ITGB1. Western blotting confirmed that transfection of miRNA-29a-3p and miRNA-29b-3p mimetics significantly reduced ITGB1 protein expression (Extended Data Fig. 4), further supporting a direct regulatory effect of the miRNA-29 family on integrin signaling.

We then optimized and generated engineered EVs overexpressing miRNA-29a-3p and miRNA-29b-3p to recapitulate this regulatory effect in an EV-mediated delivery context (Fig. 7b). Engineered EVs significantly suppressed ITGB1 expression in recipient chondrocytes, which was validated by western blotting and immunofluorescence (Fig. 7b and Extended Data Fig. 5). In parallel, treatment with miRNA-29-enriched engineered EVs increased α-Klotho expression in aged human chondrocytes (Fig. 7c). Subsequently, treatment with engineered EVs significantly increased the expression of key chondrogenic markers, including type II collagen and aggrecan, as well as a catabolic marker, MMP13, thereby recapitulating the pro-anabolic phenotype induced by exercise-primed EVs (Fig. 7d). To determine whether α-Klotho mediates these effects, we silenced Klotho expression before engineered EV treatment. Klotho knockdown partially attenuated the miRNA-29-driven upregulation of type II collagen and aggrecan, as well as downregulation of MMP13 (Fig. 7e). Taken together, these data provide mechanistic validation of our network-based predictions and suggest that miRNA-29 acts as a dominant molecular effector linking aerobic exercise to improved chondrocyte health.

Discussion

Although EVs have been recognized as effective tools to harness the systemic benefits of exercise, mechanistic evidence that exercise enhances the accumulation of health-promoting biomolecules in EVs in the context of aging has been lacking. This knowledge gap is particularly important for age-related KOA because current treatments focus primarily on symptom relief. To this end, this study introduced a network-paradigm approach to infer functional targets of exercise-primed EVs in recipient aged chondrocytes. This analysis predicted that exercise-primed EVs target chondrocyte–matrix adhesion and highlighted the miRNA-29 family as a dominant regulator of these downstream effects (see graphical abstract; Extended Data Fig. 6). Consistent with these predictions, EVs collected after exercise restored healthy chondrocyte morphology and anabolic markers in cells cultured on stiff ECM, a typical feature of aged articular cartilage[28]. These findings were further supported by in vivo evidence demonstrating that administration of exercise-primed EVs restored cartilage integrity in aged joints, highlighting their functional relevance at the tissue level. The youthful phenotype induced by postexercise EVs was mediated, at least in part, by epigenetic derepression of Klotho. Network-perturbation analysis identified the miRNA-29 family as a dominant regulator of exercise-induced EV signaling, and engineered EVs enriched with miRNA-29 family members recapitulated these anabolic effects. Partial attenuation of these effects following Klotho knockdown establishes the miRNA-29–Klotho axis as a central

https://doi.org/10.1038/s43587-026-01225-9

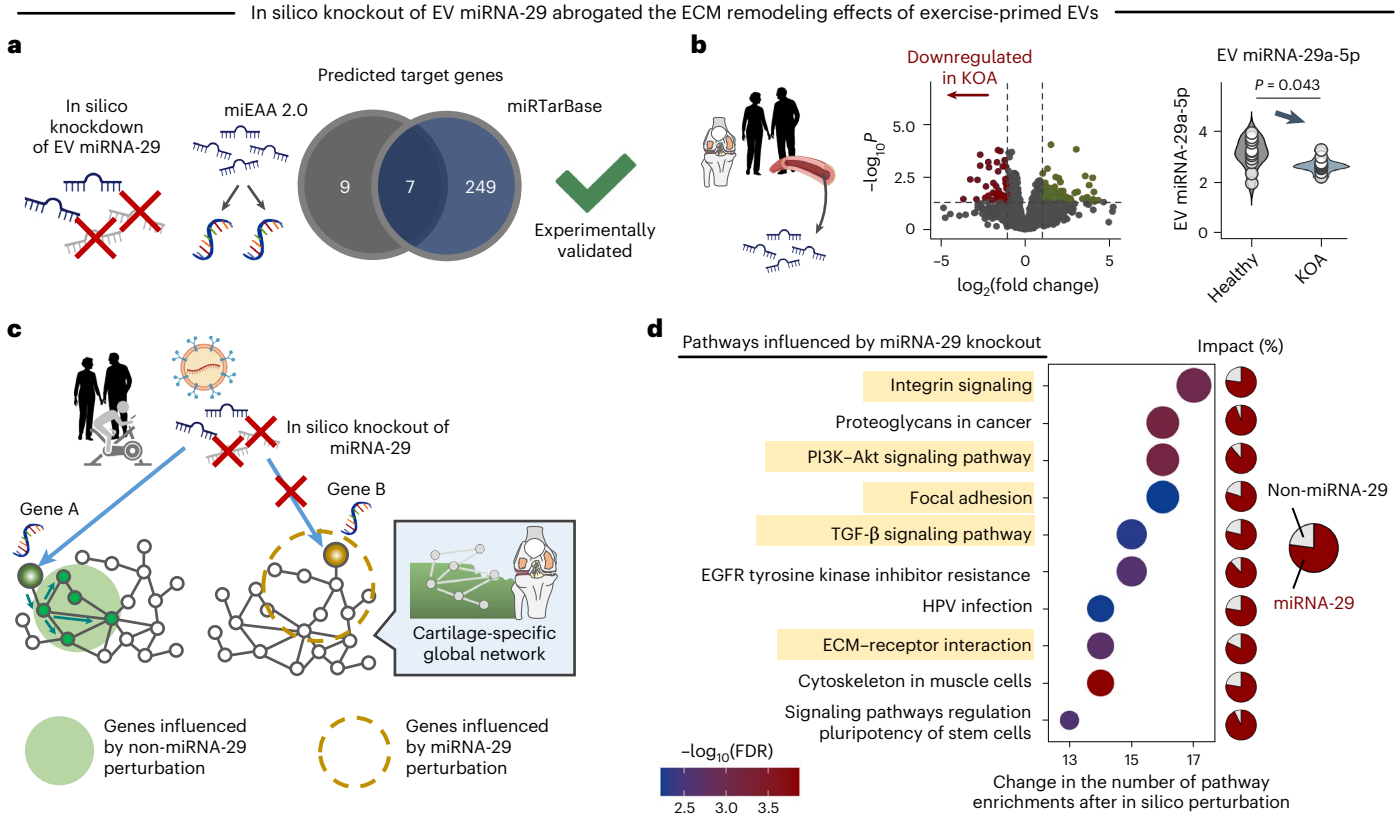


Fig. 6 | In silico knockout of miRNA-29 attenuated the health-promoting effects of exercise-primed EVs, identifying miRNA-29 as a key regulator of cartilage aging. a, Predicted EV miRNA target genes identified using miEAA 2.0 and miRTarBase, following in silico deletion of miRNA-29, demonstrating a marked reduction in experimentally validated targets. **b**, Volcano plot showing circulating miRNAs and downregulated miR-29a-5p levels in individuals with KOA compared with healthy counterparts ($n = 10$ for healthy, $n = 12$ for KOA) (GSE105027)[88]. Differentially expressed miRNAs were identified using an unadjusted nominal two-sided P value threshold of $-\log_{10}(P) > 1.3$ and $|\log_2(\text{fold change})| > 1.0$. **c**, Schematic of network perturbation, illustrating the effects of in silico miRNA-29 deletion on the cartilage-specific interaction network. **d**, Pathway enrichment analysis following in silico miRNA-29 knockout, highlighting key age-related KOA pathways. Pie charts denote the proportion of pathway impact attributable to miRNA-29 versus non-miRNA-29 miRNAs. **e**, Digital PCR quantification of EV miRNA-29a-3p and miRNA-29b-3p levels of EV^{pre} and EV^{post} ($n = 10$ per group). RFU, relative fluorescence unit. **f**, Positive correlations between exercise-induced changes in EV miRNA-29a-3p and miRNA-29b-3p and corresponding increases in α-Klotho expression in aged chondrocytes cultured on stiff substrate ($n = 10$ per group). The black line indicates the fitted linear regression line, and the shaded gray area represents the 95% confidence interval of the estimated regression line. The α-Klotho data shown are reproduced from Fig. 4d. Inter-individual variability in miRNA-29 responses supports its potential utility as a biomarker to distinguish responders from non-responders to exercise-induced cartilage-protective signaling. Statistical analyses were performed using the R package limma based on a two-sided empirical Bayes-moderated t-test as well as a two-tailed Student's t-test (**b**), a two-tailed paired t-test (**e**) and linear regression (**f**). The yellow highlighted pathways represent signaling previously implicated in the pathogenesis of age-related KOA[25]. All raw data supporting these findings are available in the Supplementary Data. Data visualizations created in BioRender; Iijima, H. http://BioRender.com/f55bqvo (2026).

mediator of EV-induced cartilage protection, highlighting potential applications for biomarker-guided and EV-based therapies for age-related KOA.

The significance of the network paradigm in this study is the inference of downstream targets of EV miRNAs in a tissue-specific manner. Unlike conventional miRNA-target prediction, which focuses on direct targets, network propagation captures broader cellular responses. For example, COL2A1 is not a direct target of miRNA-29a, but levels of this protein have been shown to increase following miRNA-29a transfection[53]. By implementing the network-propagation approach

https://doi.org/10.1038/s43587-026-01225-9

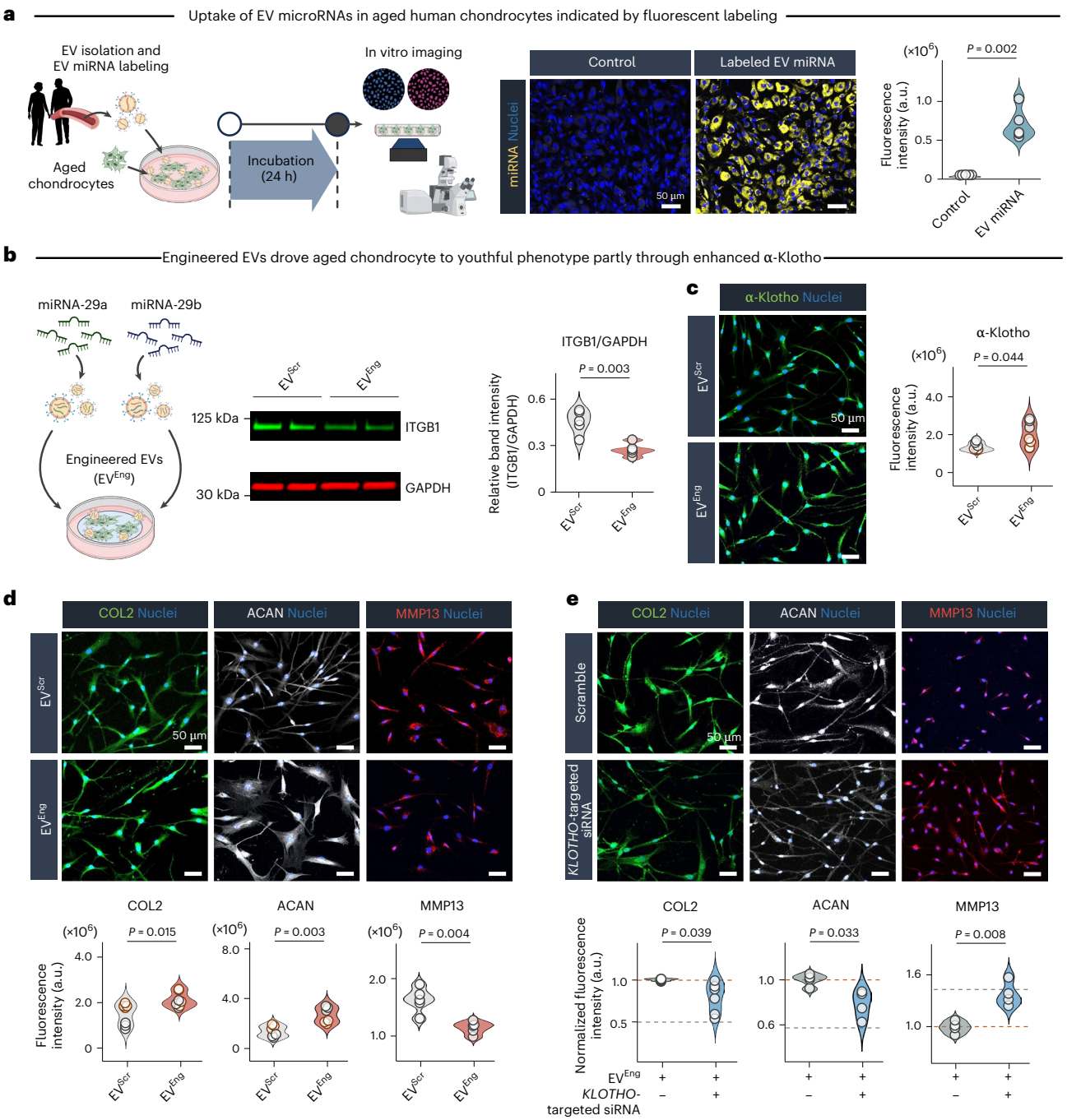


Fig. 7 | Engineered EVs enriched with miRNA-29 restore anabolic phenotype in aged human chondrocytes through partial dependence on α-Klotho. **a**, Schematic of EV miRNA fluorescent labeling, followed by incubation with aged human chondrocytes. Representative fluorescence images demonstrate efficient internalization of EV miRNAs (yellow) into chondrocytes. Quantification of fluorescence intensity confirms significant uptake of labeled EV miRNAs compared with control ($n = 5$ per group; 200–500 cells quantified per individual sample). **b**, Treatment with engineered miRNA-29-enriched EVs significantly suppressed ITGB1 expression in chondrocytes compared with scramble counterparts, as validated by western blot ($n = 5$ per group). Samples shown in each blot were derived from the same experiment and processed in parallel. **c**, Increased α-Klotho expression in aged human chondrocytes treated with engineered EVs compared with scramble EV counterparts ($n = 7$ per group; 300–1,000 cells quantified per individual sample). Data from different cohorts or experimental groups performed on different days are presented within the same graph as black or orange circles. **d**, Engineered EV treatment increased expression of anabolic chondrogenic markers, including type II collagen, aggrecan and the catabolic marker MMP13 compared with scramble EV counterparts ($n = 7$ per group for type II collagen and aggrecan, $n = 5$ per group for MMP13; 300–1,000 cells quantified per individual sample). Data from different cohorts or experimental groups performed on different days are presented within the same graph as black or orange circles. **e**, siRNA-mediated knockdown of *KLOTHO* prior to engineered EV treatment partially attenuated the miRNA-29-induced upregulation of type II collagen and aggrecan, as well as the downregulation of MMP13 ($n = 6$ per group for type II collagen, $n = 4$ per group for aggrecan, $n = 4$ per group for MMP13; 300–1,000 cells quantified per individual sample). Data from different cohorts or experimental groups performed on different days are presented within the same graph as black or orange circles. Statistical analyses were performed using two-tailed Student's *t*-tests (**a**–**e**). For violin plots, values obtained from chondrocytes cultured on stiff substrates with EV^{scr} (gray dotted line) and EV^{eng} (orange dotted line) are shown as reference lines. Data visualizations created in BioRender; Iijima, H. http://BioRender.com/f55bqvo (2026).

on a tissue-specific network, we identified pathways such as focal adhesion and PI3K–Akt signaling as downstream mediators of exercise-modulated EV miRNAs, demonstrating how this paradigm can uncover mechanistically relevant effects on chondrocyte aging.

Building on the in silico network-based inference, we demonstrated that exercise-primed EVs abrogated the deleterious effects of a stiff matrix on chondrocyte aging. It is well established that an age-related increase in matrix stiffness disrupts chondrocyte function through mechanotransductive pathways[30–32]. Our recent study showed that a stiff matrix epigenetically represses the gene encoding the longevity factor α-Klotho, resulting in chondrocyte dysfunction[25]. The current study provides evidence that exercise-primed EVs promote a youthful phenotype in aged chondrocytes under stiff matrix conditions, as evidenced by improved chondrogenicity and epigenetic reprogramming of Klotho.

Our network-perturbation analyses identified the miRNA-29 family as a dominant regulatory component underlying the chondroprotective effects of exercise-primed EVs. Engineered EVs enriched with miRNA-29a-3p and miRNA-29b-3p recapitulated the anabolic and rejuvenating effects of exercise-derived EVs in aged human chondrocytes, with partial dependence on the miRNA-29–Klotho axis. Engineered delivery of miRNA-29 through EVs has shown therapeutic efficacy in multiple disease contexts, including attenuation of skeletal muscle atrophy and kidney fibrosis, and reduction of β-amyloid accumulation in murine models of Alzheimer's disease[54–56]. Our findings extend this therapeutic paradigm to age-related KOA, a condition for which no established disease-modifying treatments are currently available. The results therefore highlight the potential of EV-based strategies as biomarker-guided, disease-modifying approaches for degenerative joint disease. Notably, the beneficial effects of exercise-primed EVs extended beyond in vitro observations, considering that intra-articular administration of EVs attenuated cartilage degeneration in aged mice. Given that no therapies approved by the US Food and Drug Administration target ECM stiffness or its downstream cellular consequences, these results highlight the therapeutic potential of EV-based approaches within the emerging framework of mechanomedicine[57].

Although this study provides mechanistic insight into geroprotective effects of exercise, it has limitations. First, it did not address how exercise alters EV bioactivity. It is reasonable to hypothesize that mechanical stress imposed by exercise alters the biogenesis and bioactivity of EVs, and this is supported by studies demonstrating that mechanical stimulation of mesenchymal stem cells and lung epithelial cells promoted EV release and the expression of miRNAs in EVs[58,59]. Further research is needed to establish the mechanism through which exercise promotes the bioactivity of circulating EVs. Similarly, the tissue sources of circulating EVs released in response to exercise remain unclear and are a worthwhile area of investigation. Second, the experimental study that compared the effects of circulating EV^{pre} versus EV^{post} was based on older adults. As such, the findings might not be generalizable to other age groups. Third, the exercise modality used in the current study was limited to aerobic exercise. As such, it is unclear whether the findings of this study are applicable to other types of exercise (for example, resistance exercise). Finally, our study focused on cartilage as a representative tissue, and the conclusions are likely to be context-specific. Importantly, although our findings suggest that circulating EVs can influence chondrocyte biology, we did not directly establish the extent to which circulating EVs penetrate human articular cartilage tissue. Cartilage is an avascular tissue whose homeostasis critically depends on the local microenvironment, including interactions between chondrocytes and ECM. By using a cartilage-specific gene regulatory network in our in silico analyses, we modeled transcriptional responses of chondrocytes to exercise-primed EV miRNAs in a manner that is directly relevant to the unique biology of cartilage. While it remains uncertain whether similar responses occur in other tissues or disease contexts, the network-based framework offers the possibility to apply a similar approach elsewhere by using appropriate tissue-specific networks. This highlights both the tissue-specific relevance of our current findings and the potential, though not yet demonstrated, for broader application of this strategy to investigate EV-mediated signaling in other organ systems.

Methods

Systematic literature search

We performed a systematic search to identify original articles investigating profiles of biomolecules in EVs (that is, proteomics, RNA-seq, small RNA-seq, microarray) in older adults after aerobic exercise. Two independent reviewers (H.I. and E.D.) with expertise in both the scientific content and systematic review methodology conducted an electronic search from the time of database inception to December 2021 using PubMed, Scopus and Web of Science. To ensure the inclusion of the most recent evidence, we subsequently conducted an updated systematic literature search in January 2026 using the same databases, search strategy and inclusion criteria, in accordance with our previously published standard systematic review flow[44,60]. A manual search was also performed using Google Scholar. The reference lists of relevant systematic reviews were manually searched[1,6,37,61–64]. The same two reviewers then assessed the titles and abstracts of records identified using the prespecified eligibility criteria. Full manuscripts of articles that passed this initial screening were then reviewed in detail. The exclusion criteria included non-peer-reviewed journals, non-original articles, languages other than English, animal studies, participants younger than 65, no aerobic exercise intervention, no circulating EVs, no omics data, unavailability of full text and unavailability of raw data. Disagreements regarding manuscript inclusion between the two reviewers were discussed until a consensus was achieved. Finally, a citation search of the identified articles was performed using Web of Science to screen for potential articles. Identified records were then screened and reviewed using the same criteria. Customized Google spreadsheets were used by reviewers to assess and record eligibility.

Differential expression analysis for archived EV small RNA-seq data

We accessed the archived small RNA-seq data (GSE144627) collected from serum in male older adults ($n = 10$; mean age, 69.3 years) at baseline (pre-exercise), immediately after and 3 h following an acute bout of aerobic exercise[3]. All participants completed a single session of moderate-intensity aerobic exercise on a cycle ergometer (~70% heart rate reserve for 40 min). Data preprocessing and differential gene expression were performed in accordance with the established workflow[65]. Raw count data were normalized by counts per million. We removed genes that had very low counts before downstream analysis on biological and statistical grounds[66]. From a statistical point of view, genes with consistently low counts or low signal intensity are very unlikely to be assessed as differentially expressed because low counts or low signal intensity do not provide enough statistical evidence for a reliable judgement to be made. Such genes can therefore be removed from the analysis without any loss of information. We used the filterByExpr function of the edgeR R package, with default parameters for low count data (min count = 10)[67]. The count data were further normalized using trimmed mean of *M* values normalization and voom transformation using the edgeR and limma R packages[67,68]. The differential gene expression analysis was performed using the limma R package[69].

Target gene prediction of miRNAs

Primary target genes of differentially expressed miRNAs were determined using two different databases, miEAA.2.0 and miRTarBase[17,18]. miEAA.2.0 is an updated web-based application that offers a variety of commonly applied statistical tests such as over-representation analysis and miRNA set enrichment analysis[17]. The target genes significantly predicted by mmiEAA.2.0 were then cross-checked by miRTarBase, which is a database of experimentally validated miRNA–target

interactions[18]. Among the experimentally validated miRNAs, we have included genes with 'strong' experimental evidence (that is, genes supported by reporter assay, western blot or qRT–PCR).

Functional characterization of target genes using GO enrichment analysis on a cartilage-specific functional network

To characterize the biological function of genes of interest, GO enrichment analysis was performed on the cartilage-specific functional gene network generated with Genome-wide Integrated Analysis of gene Networks in Tissues (GIANT 2.0)[10]. GIANT 2.0 predicts tissue- and cell-type-specific functional interactions by probabilistically integrating large-scale functional genomics datasets, enabling inference of coordinated biological processes in contexts in which direct experimental interrogation is not feasible[10]. Because GIANT 2.0 networks are derived from human tissue and cell-type-specific data, all gene symbols were converted to human gene symbols prior to analysis, consistent with previous work[70].

Gene perturbation on miRNA regulatory network using RWR

A cartilage-specific functional network was constructed using HumanBase, based on curated gold-standard gene interaction data[21]. The network was first converted from Entrez ID nos. to human gene symbols to enable downstream analyses. RWR was then performed on this network using the R/Bioconductor package RandomWalkRestartMH[24]. Each of the EV miRNA-target genes was used as a single seed node, and the RWR algorithm propagated influence through the network with a global restart probability of $r = 0.7$, following default parameter settings established in previous studies[24,71–74]. This process was repeated individually for each seed gene, generating per-seed affinity scores representing the proximity of all nodes to the corresponding seed gene.

Functional characterization of in-silico-activated genes using pathway enrichment analysis

To determine the functional consequences of gene perturbation on the miRNA regulatory network, we performed KEGG pathway enrichment analysis using the R package clusterProfiler[75]. For each seed gene, the top 50 pseudo-activated genes ranked by affinity score were selected as input for enrichment. Gene symbols were converted to Entrez IDs prior to analysis. Enrichment was performed using the Benjamini–Hochberg procedure for multiple testing correction, with a q-value cutoff of 0.05.

In silico miRNA-29 knockout

Building on the KEGG enrichment results obtained from the pseudo-activated gene sets described above, we next quantified the specific contribution of the miRNA-29 family at the pathway level using an in silico knockout strategy. KEGG enrichment outputs were generated separately for (1) the complete miRNA set (all miRNAs) and (2) a dataset in which miRNA-29 family members were computationally removed. For each condition, enriched pathways across all seed genes were aggregated, and the number of times each KEGG pathway appeared (seed-based enrichment frequency) was calculated. To quantify miRNA-29-dependent effects, an impact score was defined as the difference in pathway frequency between the two conditions:

$$\text{Impact score}_i = \text{Count}_{\text{all miRNA},i} - \text{Count}_{\text{miRNA-29knockout},i} \quad (1)$$

where i denotes an individual KEGG pathway. A positive impact score indicates pathways preferentially driven by miRNA-29. To normalize for baseline pathway frequency, percentage impact was calculated as the proportion of the impact score relative to the total miRNA count.

Human chondrocytes culture

This study followed the BRISQ reporting guideline[76]. Primary human chondrocytes from a healthy older male donor (65 years; BMI, 23.4 kg m^{-2}; donor ID no. HECA-20-12.0) were obtained through the StemBioSys company. Isolated chondrocytes were seeded in tissue-culture flasks at a density of 0.5×10^4 cells cm^{-2} and maintained in low-glucose DMEM supplemented with 15% FBS, 1% GlutaMAX and 1% penicillin–streptomycin at 37 °C at 5% CO_2. After cells were fully adhered to the culture substrate, the medium was changed every 3 days until cells reached 70–80% confluence. The cells were then detached with Trypsin/EDTA (Gibco/Thermo Fisher Scientific) and passaged. For all experiments, second-passage chondrocytes were used. In preliminary studies, in vitro treatments with EV^{pre} and EV^{post} were also performed using a second, independent source of commercially obtained human chondrocytes (male donor, 68 years old; donor ID no. CD-20-02.0; StemBioSys), which had been used in our prior study[46]. Results from the donor were similar to those obtained from the 65-year-old donor, particularly for key anabolic markers including type II collagen and SOX9. On the basis of this reproducibility, all subsequent experiments were conducted using chondrocytes from the 65-year-old donor to ensure experimental consistency.

Aerobic exercise cohort

The aerobic exercise cohort was derived from the Cognition and Exercise (CogEx) trial (ClinicalTrials.gov identifier: NCT03696082), a randomized controlled trial designed to investigate the effects of exercise interventions in community-dwelling older adults. The study protocol was reviewed and approved by the Institutional Review Board of the University of Pittsburgh, and all participants provided written informed consent before enrollment and study procedures. Participants were recruited from the community; the primary recruitment method was a clinical registry of individuals who expressed interest in participating in research studies. Interested individuals completed a telephone screening to determine initial eligibility. Potentially eligible individuals attended a study orientation session, during which additional details of the study were provided, and written informed consent was obtained prior to proceeding with additional study procedures.

Eligible participants were community-dwelling older adults aged 65–80 years who met the following inclusion criteria: (1) sedentary status, defined as engaging in <60 min per week of moderate-to-vigorous leisure-time or transportation-related physical activity performed in bouts of >10 minutes over the past three months; (2) independent ambulation, with or without the use of a cane or walking stick; (3) ability to walk at least 400 m, based on self-report; (4) physician clearance for participation in study assessments and the intervention; and (5) ability to provide informed consent. Participants were excluded if they met any of the following criteria: (1) regular participation in aerobic exercise, resistance training or yoga during the preceding three months; (2) medical conditions contraindicating moderate-intensity physical activity, including stroke, uncontrolled hypertension, congestive heart failure or recent myocardial infarction; (3) severe neurological or respiratory disorders; (4) inability to comply with the intervention schedule; or (5) other conditions that could interfere with study assessments or intervention participation (for example, severe cognitive impairment, uncontrolled depressive symptoms or use of assistive devices beyond a cane or walking stick). All participants were required to obtain physician clearance before participation in the assessments and intervention. Participants did not receive financial or other compensation for participation in the study.

Participants who remained eligible were assigned to participate in the intervention, with one of the intervention conditions being aerobic exercise ($n = 10$; mean age, 69.4 ± 3.7 years; 80% female participants). Detailed participant characteristics are provided in Supplementary Table 2. At baseline, participants in the aerobic exercise group had a mean body weight of 83.2 ± 15.9 kg and a mean BMI of 30.7 ± 6.4 kg m^{-2}. Following the 3-month intervention, body weight (83.4 ± 16.6 kg; change, 0.2 ± 2.4 kg; $P = 0.830$) and BMI (30.9 ± 7.0 kg m^{-2}; change, 0.2 ± 0.9 kg m^{-2}; $P = 0.679$) did not significantly change from baseline. The EV analyses reported in this paper were exploratory

analyses using biospecimens collected during the CogEx trial and were not prespecified registered trial outcomes. Serum samples used for EV isolation were collected according to the original study protocol and were obtained following completion of the 12-week intervention and prior to trial termination.

The CogEx trial was subsequently terminated because institutional funding support for the study was discontinued. This termination was an administrative and funding-related decision and was not related to adverse events, participant safety concerns or issues involving individual participants. Importantly, none of the participants whose EV samples were analyzed in this study were associated with the reason for trial termination. Although the 12-week exercise intervention and biospecimen collection were completed as originally planned, the trial was terminated before completion of the planned long-term follow-up assessments. Therefore, functional outcome data from these follow-up assessments were not collected and are not reported.

The aerobic exercise intervention was designed as a real-world, hybrid program combining supervised and home-based sessions. Participants completed one supervised session and four home-based sessions per week. Supervised sessions lasted 30 min each, and home-based sessions progressed from 10 min per session in weeks 1–4, to 20 min in weeks 5–8, and 30 min in weeks 9–12, resulting in a total weekly exercise duration of 70–150 min. Exercise intensity was guided by participants' perceived exertion. Participants were encouraged to maintain a level they subjectively considered moderate intensity, reflecting practical, community-based exercise conditions. To facilitate this, aerobic exercise videos were developed by the study staff, and these were used to guide the supervised exercise sessions and were placed on tablets provided to the participants to facilitate the unsupervised, home-based exercise sessions. However, the participants could also substitute other forms of aerobic exercise (for example, walking) for the unsupervised, home-based sessions if they chose to do so, but they were instructed not to initiate resistance exercise.

Serum collection

Blood was collected by trained phlebotomists in the morning between 7:30 a.m. and 10:30 a.m. through venipuncture using 21-gauge needles, with collection timing depending on participant availability. To minimize pre-analytical variability in circulating EVs, participants were instructed to fast for 10–12 h, except for water, and to avoid strenuous physical activity for 10–12 h prior to blood collection. Fasting was implemented to control for potential effects of recent food intake, and adherence to these instructions was confirmed at the study visit. Blood for serum preparation was collected into red-top serum tubes without anticoagulant after an initial discard tube. Serum tubes were maintained upright (were not inverted) during and after collection. Following venipuncture, samples were allowed to clot undisturbed for 30–45 min at room temperature and were processed within 2 h of blood collection. Serum tubes were centrifuged at 2,500 r.p.m. for 20 min at room temperature. The serum supernatant was carefully aliquoted into sterile 500-µl cryovials to minimize repeated freeze–thaw cycles and stored at −80°C until EV isolation. Samples underwent a single freeze–thaw cycle before EV isolation. Samples exhibiting visible hemolysis, identified by pink or red discoloration upon visual inspection, were excluded from downstream analyses. All samples were collected, processed and stored according to a standardized protocol to minimize pre-analytical variability, consistent with the MIBlood-EV recommendations[77].

EV isolation and characterization

EVs were isolated from serum using size-exclusion chromatography (cat. no., SP6, qEVsingle 35-nm iZON columns) according to the manufacturer's protocol and as previously described[35]. Following column equilibration with PBS, serum samples were applied to the column and sequential fractions were collected. Fractions 1–5 were collected as the void fractions and defined as the non-EV fraction, as they predominantly contain soluble proteins and minimal EV content according to the manufacturer's guidelines. Fractions 6–11 were subsequently collected and pooled as the EV-enriched fraction (EV fraction). Both non-EV and EV fractions were isolated separately and processed independently for downstream analyses. Isolated fractions were stored at −20°C until further use. EV characterization was performed using complementary approaches in accordance with the MISEV recommendations[36,78], including quantitative, molecular, biophysical and functional analyses (Supplementary Table 3), as described below.

Nanoparticle tracking analysis of EVs

EVs were characterized for particle size distribution and concentration using nanoparticle tracking analysis (NTA) on a NanoSight LM10 instrument (Malvern Instruments) equipped with an sCMOS camera and a 405-nm blue laser. Measurements were performed at 22 °C. EV samples (10 µl) were diluted 1:100 in freshly prepared 0.22-µm-filtered PBS (pH 7.4) and introduced into the flow cell using a syringe pump. For each sample, three videos were recorded with the camera level set to 7 and gain set to 15. Videos were analyzed in batches using NTA software (v3.4) with a detection threshold of 2. Filtered PBS was analyzed as a blank prior to EV measurements, and the flow cell was washed with 1 ml filtered PBS between samples to prevent carryover.

Protein-based EV characterization and purity assessment

EV protein markers and sample purity were assessed by western blot analysis. Human aged male serum (150 µl) was loaded onto qEVsingle 35-nm size-exclusion chromatography columns (Izon Science), and non-EV and EV fractions (1.25 ml each) were collected, according to the manufacturer's protocol. To assess albumin enrichment relative to whole serum, an additional 150 µl serum sample was diluted in 2.5 ml PBS. Collected fractions were transferred to polycarbonate tubes and ultracentrifuged at 120,000*g* for 80 min at 4 °C using a Beckman Coulter ultracentrifuge. Supernatants were discarded, and pellets were resuspended in an equal volume of PBS. The total protein concentration was quantified using a Pierce bicinchoninic acid (BCA) Protein Assay Kits (Fisher Scientific), according to the manufacturer's instructions.

For EV protein characterization, 10 µg total protein from lysed EV fractions was mixed with Laemmli sample buffer and denatured at 95 °C for 5 min. Samples and molecular weight standards (8–250 kDa) were separated on 4–20% Tris-Glycine polyacrylamide gels (Thermo Fisher Scientific) at 150 V for 40 min and transferred onto 0.45-µm PVDF membranes (Millipore Sigma) at 65 V for 90 min using a Mini Trans-Blot system (ThermoFisher Scientific). Membranes were washed three times for 10 min in Tris-buffered saline containing 0.1% Tween-20 (TBS-T; pH 7.6) and blocked with Intercept blocking buffer (LICOR-bio; cat. no. 927-60001) for 1 h at room temperature. Membranes were incubated overnight at 4 °C with primary antibodies to CD9 (2 µg ml^{-1}, Thermo Fisher Scientific; cat. no. 10626D), CD63 (1:1,000, Santa Cruz Biotechnology; cat. no. sc-5275), CD81 (1:1,000, Invitrogen; cat. no. MA5-13548), calnexin (1:1,000, Invitrogen; cat. no. MA3-027) and albumin (1:1,000, Thermo Fisher Scientific; cat. no. 16475-1-AP) diluted in blocking buffer. After washing (three times, 10 min each), membranes were incubated with IRDye 680RD-(LICOR; cat. no. 926-68070) and 800RD-conjugated (LICOR; cat. no. 926-32211) secondary antibodies (1:10,000 dilution) for 1 h at room temperature, followed by additional washes in TBS-T. Blots were imaged at 700- and 800-nm channels using an Odyssey Classic imaging system (LICOR), and band intensities were quantified using Image Studio software (v5.2). For EV purity assessment, 30 µl of ultracentrifuged serum, non-EV fractions and EV fractions were processed for western blotting as described above and evaluated for albumin enrichment. All experiments were performed in triplicate using three independent serum samples.

Quantification of EV miRNA-29 level

Total RNA was extracted from human serum-derived EVs (200 µl EV fraction) using the miRNeasy Serum/Plasma Advanced Kit (Qiagen, cat. no., 217204), according to the manufacturer's instructions. Owing to the low RNA yield typical of biofluid-derived EV samples, RNA input into the reverse-transcription reaction was calculated on the basis of the original serum volume. Complementary DNA synthesis was performed using the miRCURY LNA RT Kit (Qiagen, cat. no. 339340), which employs poly(A) tailing followed by universal reverse transcription. RT reactions were incubated at 42 °C for 60 min and subsequently heat-inactivated at 95°C for 5 min. The resulting cDNA was diluted before digital PCR analysis. Absolute quantification of hsa-miR-29a-3p and hsa-miR-29b-3p was carried out using miRNA-specific LNA-enhanced assays (miRCURY LNA miRNA PCR Assay) in combination with the QIAcuity EG PCR Kit on the QIAcuity Digital PCR System (Qiagen). Digital PCR was performed according to the manufacturer's protocol, with an initial polymerase activation step, followed by 40 cycles of denaturation and annealing and extension. EvaGreen fluorescence was detected in the green channel. Absolute copy numbers were calculated using the QIAcuity Software Suite on the basis of positive/negative partition discrimination and Poisson statistics.

Preparation of fibronectin-coated polyacrylamide substrates

We prepared polyacrylamide (pAAm) gels with different stiffnesses (5 kPa and 100 kPa) in accordance with a previous study[79]. The pAAm gels were made on glass coverslips pre-treated with 0.1 N sodium hydroxide (cat. no. SS255-1, Fisher Scientific), 0.5% 3-aminopropyltrimethoxysilane (cat. no. AC313251000, Acros Organics) and 0.5% glutaraldehyde (cat. no. BP25481, Fisher Scientific) to improve gel adhesion. To facilitate cell adhesion, the surfaces of prepared hydrogels were further conjugated with fibronectin (100 µg ml^{-1}, from bovine plasma, Sigma) using sulfo-SANPAH (cat. no. NC1314883, Proteochem) as a cross-linker. Before seeding cells, gels were ultraviolet-sterilized in a cell-culture hood for 30 min. Gels were kept hydrated in HEPES or PBS during all preparation steps.

Evaluation of the impact of matrix stiffness on chondrocyte phenotype

Primary human chondrocytes isolated from an older male donor (65 years old) were plated at a density of 5,000 cells cm^{-2} onto pAAm gels with elastic moduli of 5 kPa (young-like) or 100 kPa (aged-like). Cells were cultured in low-glucose DMEM supplemented with 15% fetal bovine serum (FBS), 1% GlutaMAX, and 1% penicillin/streptomycin at 37°C in 5% CO_2. On day 5, culture medium was removed and cell-seeded pAAm gels were fixed with 2% paraformaldehyde (PFA) for 10 minutes. After three washes with PBS, samples were stored in PBS at 4 °C until further processing.

Assessment of the therapeutic potential of circulating EVs

To test the ability of circulating EVs isolated before and after exercise to modulate chondrocyte health, the isolated EVs were administered (0.5×10^9 particles ml^{-1}) to aged human chondrocytes cultured on aged-like stiff pAAm substrate on day 3 after chondrocyte seeding. After an additional 2 days of culture with EVs (that is, on day 5 after chondrocyte seeding), culture medium was removed and samples were fixed in 2% PFA for 10 min. After three washes with PBS, samples were stored in PBS at 4 °C until further processing.

Immunofluorescence and imaging

Immunofluorescence analyses were performed to quantify expression of type II collagen, aggrecan, SOX9, MMP13, ADAMTS4, ITGB1, DNMT1 and α-Klotho using established protocols[80]. In brief, for cultured chondrocytes, samples were washed three times with PBS, permeabilized with 0.1% Triton X-100 for 15 min, and blocked for 1 h in PBS containing 0.1% Triton X-100 (Fluka 93420) and 3% bovine serum albumin (BSA; Sigma A7906). For tissue sections, paraffin-embedded slides were dried overnight at 60 °C, deparaffinized and rehydrated through a graded ethanol series (100%, 95% and 70%), followed by PBS. Antigen retrieval was performed in sodium citrate buffer using a water bath at 92 °C for 10 min. Slides were allowed to cool to room temperature, washed three times with PBS and blocked for 1 h in blocking buffer (PBS containing 0.1% Triton X-100 and 3% BSA). Samples were incubated overnight at 4°C with primary antibodies to type II collagen (1:200; ab34712, Abcam), aggrecan (1:200; MA3-16888, Invitrogen), SOX9 (1:200; cat. no. 82630, Cell Signaling), MMP13 (1:200; ab39012, Abcam), ADAMTS4 (1:200; ab185722, Abcam), ITGB1 (1:200; ab179471, Abcam), DNMT1 (1:200; no. 5032, Cell Signaling) and α-Klotho (1:200; 600-401-CF4, Rockland), diluted in blocking buffer supplemented with 5% goat serum. Negative control samples were prepared for each staining set by omitting the primary antibody. After three PBS washes, samples were incubated with host-specific Alexa-Fluor-488-conjugated secondary antibodies (1:200; Fisher Scientific) in blocking buffer for 1 h at room temperature, followed by DAPI nuclear staining for 2 min. Samples were mounted with coverslips using mounting medium (VectaMount AQ Aqueous Mounting Medium; H-5501-60-NB, Vector Labs). Images were acquired using a Nikon AXR point-scanning confocal microscope equipped with a 20× objective. Negative control slides were used to determine the threshold for signal intensity and to set the exposure time for individual channels. All images for quantitative analysis in a given experiment were taken under the same imaging conditions. Fluorescence intensity was quantified using ImageJ, in which the integrated density was divided by the number of cells in each image. We took 5–10 random images in each sample and averaged them for statistical analysis.

Quantification of cellular morphology

F-actin was visualized using phalloidin co-incubated with secondary antibodies during immunofluorescence staining described above. Following primary antibody incubation and three PBS washes, samples were incubated for 1 hour at room temperature with host-specific Alexa Fluor 488–conjugated secondary antibodies mixed with Alexa Fluor 594–conjugated phalloidin (1:200; Invitrogen; cat. no. R415) in blocking buffer. Nuclei were counterstained with DAPI for 2 minutes, and samples were mounted with VectaMount® AQ medium. Images were captured using the Nikon AXR confocal microscope at 20× magnification under the same imaging conditions as the immunofluorescence analysis.

For quantitative analysis of cellular morphology, F-actin images were processed using CellProfiler software (v4.2.6, The Broad Institute)[38]. Cells were identified using the 'IdentifyPrimaryObjects" module, and 53 morphological features were extracted using the 'MeasureObjectSizeShape' module. Data was exported to spreadsheets for downstream analysis. For each sample, 5–10 images were acquired and averaged for statistical comparisons.

EV uptake in cultured human chondrocytes

EV uptake was assessed by detecting fluorescence signals from labeled EVs internalized by primary human chondrocytes cultured on pAAm hydrogels of defined stiffness (soft or stiff). For EV labeling, vesicles (1×10^{10} particles ml^{-1}) were incubated with 10 µM calcein-AM (BD Pharmingen) for 30 min at 37 °C in the dark under gentle agitation. Following incubation, excess free dye was removed using ExoQuick exosome precipitation solution (System Biosciences) according to the manufacturer's protocol. EVs were pelleted at 1,500*g* for 30 min at 4 °C and resuspended in PBS. PBS containing calcein-AM without EVs served as a free-dye control, whereas EVs processed without calcein-AM served as unlabeled EV controls. Particle concentrations of labeled and unlabeled EV preparations were determined by NTA.

Primary human chondrocytes were seeded onto soft or stiff pAAm hydrogels and treated with labeled EV^{pre} or EV^{post} at 3.0×10^9 particles ml^{-1} for 48 h in complete growth medium at 37 °C in a

humidified incubator. Control groups received equivalent amounts of unlabeled EVs or equal volumes of free-dye control solution. After treatment, cells were washed with PBS for 5 min to remove unbound vesicles. Cells were stained with MitoTracker Deep Red (300 nM, Invitrogen) for 30 min in growth medium at 37 °C to label the cytosolic mitochondrial network, followed by PBS washing. Nuclei were stained with DAPI (1:100 dilution in PBS) for 15 min in the dark and washed with PBS prior to imaging.

Fluorescence images were acquired using a Nikon confocal microscope at 20× magnification under identical acquisition settings across all experimental conditions, with excitation at 405 nm (DAPI, nuclei), 488 nm (calcein-labeled EVs) and 647 nm (MitoTracker, cytosol). EV uptake and colocalization analyses were performed using Z-stack acquisitions and unprocessed image datasets to minimize analytical bias. Background subtraction was applied prior to quantification, and colocalization was evaluated using the Just Another Colocalization Plugin (JACoP) in FIJI software[81]. The extent of colocalization was quantified using Mander's overlap coefficient[82].

Animal handling and housing

Experiments were performed using young (4–6 months) and aged (18–24 months) male C57BL/6N mice obtained from the National Institute on Aging (NIA) Rodent Colony. These animal ages are equivalent to approximately 20–30 and 63–69 years in humans, respectively[83]. Three to five mice were housed in each cage, which were kept in a controlled environment (22–23 °C, 55–65% humidity, 12-h light/dark cycle). Food and water access was provided ad libitum. Animal usage and experiments were approved by Massachusetts General Hospital Institutional Animal Care and Use Committees (protocol ID nos. 2022N000171 and 2022N000148). Mice were not used in any experiments if they exhibited symptoms indicative of serious health concerns (for example, poor grooming, hunched behavior and/or >20% weight loss) or if they died unexpectedly before the endpoint. All mice were euthanized through CO_2 asphyxiation followed by cervical dislocation.

LI-COR imaging after systemic injection of dyed EVs in aged mice

Aged (21 months) C57BL/6N male mice were obtained from the NIA Rodent Colony. Male mice were used to maintain biological consistency with the human small RNA-seq dataset derived from an exercise cohort of aged male individuals (GSE144627)[3]. Notably, in our previous study, we did not observe age-related progressive declines in cartilage integrity in aged female mice[25], supporting the relevance of aged male mice for these analyses. Before inclusion in experiments, animals with visible health abnormalities were excluded. Mice were housed in cages holding an average of 3–4 mice per cage with a temperature-controlled environment and 12-h light/dark cycles. The animals had access to food and water ad libitum. All animal experiments were approved by the Institutional Animal Care and Use Committee. To assess biodistribution of systemically injected EVs in aged mice, EVs were labeled with PKH26 dye and imaged by LI-COR at 48 h after the systemic injection.

Ex vivo EV uptake assay

For ex vivo EV uptake assays, femurs were collected from young and aged mice and processed immediately after dissection. Surrounding soft tissues were removed, and femoral condyles were incubated with calcein-labeled EVs at 3.0×10^9 particles ml^{-1} for 48 h in complete growth medium at 37 °C in a humidified incubator. Control samples received equal volumes of free-dye control solution. Following incubation, tissues were washed thoroughly with PBS to remove unbound EVs. Femoral condyles were stained with MitoTracker Deep Red (300 nM, Invitrogen) for 30 min at 37 °C in growth medium to label the chondrocyte mitochondrial network, followed by PBS washing. Nuclei were stained with DAPI (1:100 dilution in PBS) for 15 min in the dark and washed before imaging. Fluorescence Z-stack images were acquired using a Nikon confocal microscope at 20× magnification with a step size of 1 µm under identical acquisition settings across all experimental conditions. The excitation wavelengths were 405 nm (DAPI, nuclei), 488 nm (calcein-labeled EVs) and 647 nm (MitoTracker, cytosol). EV uptake and colocalization analyses were performed using unprocessed Z-stack image datasets to minimize analytical bias. Background subtraction was applied prior to quantification, and colocalization was evaluated using the Just Another Colocalization Plugin (JACoP) in FIJI software[81]. The extent of colocalization was quantified using Mander's overlap coefficient[82].

miRNA-29 transfection and ITGB1 expression analysis

The effects of miRNA-29a-3p and miRNA-29b-3p on ITGB1 expression in human chondrocytes were evaluated by miRNA mimic transfection followed by western blot and immunofluorescence analyses. Primary human chondrocytes were seeded in 24-well plates at 5×10^4 cells per well in complete growth medium. Near confluency, cells were transfected with hsa-miRNA-29a-3p mimic (Thermo Fisher Scientific; cat. no. 4464066; assay ID MC12499) or hsa-miRNA-29b-3p mimic (cat. no. 4464066; assay ID MC10103) at 5, 25 or 50 pmol per well using Lipofectamine RNAiMAX (Thermo Fisher Scientific; cat. no. 13778075), according to the manufacturer's instructions. A mirVana miRNA mimic negative control (Thermo Fisher Scientific; cat. no. 4464058) at 25 pmol per well served as the control. In brief, miRNA mimics were complexed with RNAiMAX reagent in Opti-MEM for 5 min and subsequently diluted in complete growth medium before addition to cells. Cells were incubated with transfection complexes for 24 h at 37 °C in a humidified incubator and then washed with PBS.

For western blot analysis, cells were lysed in 1× RIPA buffer (ThermoFisher Scientific) supplemented with protease and phosphatase inhibitors. Total protein concentration was determined using a BCA assay kit according to the manufacturer's protocol. Equal amounts of protein (20 µg) were subjected to western blot analysis as described in the EV characterization section. Membranes were incubated with primary antibodies against ITGB1 (rabbit anti-ITGB1, 1:1,000, Abcam; cat. no. ab179471) and Gapdh (mouse, 1:1,000, Invitrogen; cat. no. MA-15738) as a loading control. ITGB1 band intensities were normalized to Gapdh, and relative expression levels in miR-29a- or miR-29b-transfected cells were compared with scrambled miRNA control-treated cells.

For immunofluorescence imaging, cells were fixed with 2% PFA in PBS for 15 min at room temperature and washed three times with PBS. Cells were permeabilized with 0.1% Triton X-100 in PBS for 15 min and blocked with blocking buffer (3% BSA and 0.1% Triton X-100 in PBS) for 1 h at room temperature. Cells were incubated overnight at 4 °C with rabbit anti-ITGB1 antibody (1:200 dilution) prepared in antibody dilution buffer (3% BSA, 5% goat serum, and 0.1% Triton X-100 in PBS). After washing, cells were incubated with Alexa-Fluor-488-conjugated anti-rabbit secondary antibody (1:200 dilution) for 1 h at room temperature in the dark. Nuclei were stained with DAPI (1:100 dilution), followed by PBS washing. Fluorescence images were acquired using a Nikon confocal microscope at 20× magnification under identical acquisition settings across all experimental conditions, with excitation wavelengths of 405 nm (DAPI, nuclei) and 488 nm (ITGB1).

Loading exogenous miRNA into EVs and transfer to human chondrocytes

Loading exogenous miRNA into EVs and subsequent transfer to recipient human chondrocytes were evaluated using fluorescently labeled miRNA and cellular uptake analysis. Exogenous miRNA was incorporated into EVs using a siRNA/miRNA ExoFect transfection kit (System Biosciences; cat. no. EXFT200A-1), according to the manufacturer's instructions. In brief, Cy3-dye-labeled Pre-miR negative control (10 µM; Thermo Fisher Scientific, cat. no. AM17120) was incubated with ExoFect transfection reagent in transfection buffer for 15 min at room

temperature in the dark. Serum-derived EVs (100 µg total protein) from an aged male donor were then added to the reaction mixture and incubated for 1 h at 37 °C in the dark to facilitate miRNA loading. Following incubation, samples were applied to wash columns and incubated for 15 min at room temperature to remove excess unincorporated miRNA. miRNA-loaded EVs were collected by centrifugation at 1,000*g* for 30 s into collection tubes and subsequently diluted in complete growth medium. Primary human chondrocytes were treated with miRNA-loaded EVs for 48 h. After treatment, cells were washed with PBS and fixed with 2% PFA, and nuclei were stained with DAPI (1:100 dilution in PBS). Fluorescence images were acquired using a Nikon confocal microscope at 20× magnification, with excitation wavelengths of 405 nm (DAPI, nuclei) and 594 nm (Cy3-labeled miRNA).

Engineering EVs with miRNA-29 family with and without α-Klotho inhibition

Exercise-mimetic engineered EVs were generated by loading exogenous hsa-miR-29a-3p and hsa-miR-29b-3p into serum-derived EVs from an aged male donor. miRNA loading was performed using a siRNA/miRNA ExoFect transfection kit (System Biosciences) as described above, with modified miRNA input concentrations. In brief, hsa-miRNA-29a-3p and hsa-miRNA-29b-3p mimics were individually loaded into EVs at initial amounts of 100 pmol and 10 pmol, respectively. miRNA-29a-loaded EVs and miRNA-29b-loaded EVs were prepared separately, combined at an equal volume ratio and subsequently diluted in complete growth medium prior to cell treatment. EVs loaded with mirVana miRNA mimic negative control (110 pmol initial input) served as scrambled miRNA EV controls. Primary human chondrocytes cultured on stiff pAAM hydrogels were treated with miR-29a/b-loaded EVs or scrambled control EVs for 48 h at 37 °C in a humidified incubator. Following treatment, cells were washed with PBS and subjected to immunoassays for ITGB1, collagen type II, aggrecan and MMP13 expression, as described above.

To determine whether the effects of miRNA-29-loaded EVs on chondrocyte phenotype are mediated through Klotho-dependent mechanisms, we next manipulated Klotho expression in primary human chondrocytes cultured on stiff pAAM hydrogels using siRNA-mediated knockdown. Cells were transfected with Klotho-targeting siRNA using DharmaFECT transfection reagent, according to the manufacturer's instructions. In brief, α-Klotho siRNA (5 µM) was incubated with DharmaFECT reagent in serum-free medium for 20 min at room temperature and subsequently diluted in antibiotic-free complete growth medium prior to addition to cells. Cells were incubated with transfection complexes for 6 h at 37 °C in a humidified incubator. Scrambled siRNA-treated cells served as inhibition controls. Following transfection, the reaction mixture was removed, and cells were washed with pre-warmed PBS. Cells were then treated with engineered EVs containing miR-29a and miR-29b for 48 h at 37 °C in a humidified incubator. After treatment, cells were washed with PBS and subjected to immunoassays for α-Klotho, collagen type II, aggrecan and MMP13 expression, as described above.

Intra-articular injection of EVs

Aged male C57BL/6N mice ($n = 10$) received intra-articular injections of EV^{pre} versus EV^{post}. To mitigate the effects of between-animal variability, we used a within-subjects design, as in a previous study[46]. Each mouse received intra-articular injections (two times a week) of EV^{post} into the right knee and EV^{pre} into the left knee (0.5×10^9 EVs, 10 µl per injection) for 4 weeks. EV treatment assignment was standardized across animals, with EV^{post} consistently administered to the right knee and EV^{pre} to the left knee; therefore, randomization of treatment assignment was not performed. The duration and frequency of the EV injection were determined according to previous work[35]. We developed an intra-articular knee injection protocol using a lateral injection approach to mimic that standard of care in humans[46]. In brief, each mouse knee was held in a slightly flexed position, and a 31-gauge needle was then introduced anterolaterally beneath the patellar ligament and into the intra-articular space. The knee was then cycled through flexion and extension three times, and the mouse was returned to the cage. When mice reach the appropriate end point (four weeks after the injection), mouse knee joints were collected for subsequent tissue processing and histological analyses. Body mass was similar across the start date (32.5 ± 2.6 g) and 4 weeks after the injection (31.9 ± 2.6 g). All injections were performed at the same time during the light period (8–9 a.m.). One animal was excluded from the analysis because of severe inflammation of the knee joint with loss of articular cartilage, and the remaining nine mice were used for the final analysis.

Histological preparation and semi-quantitative histological score for cartilage degeneration

Mouse knee joints were fixed in 4% PFA overnight at 4 °C and decalcified in 10% ethylene diamine tetra acetic acid solution for 10 days. Decalcified paraffin sections (5 µm thickness) were prepared from the central region of the mouse knee joints in the frontal plane in accordance with Osteoarthritis Research Society International (OARSI) recommendations[84]. The paraffin sections were stained with Safranin O/Fast Green/Hematoxylin to evaluate the severity of cartilage lesions. This study focused on medial tibial cartilage, given that this is the region most typically affected in humans, and its severity is typically equal to or higher than those of the femoral condyle in the majority of aged mice[85,86]. The OARSI scoring system, consisting of six grades and four stages on a scale from 0 (normal) to 24 (severe cartilage lesion), was used for semi-quantitative evaluation of cartilage lesion severity[87]. The most severe score for each joint was selected as the maximum OARSI score. We did not add the summed scores generated from different sections, as discriminative ability for age-related cartilage degeneration is comparable to the maximum score[86]. All slides were imaged using a Nikon AX/AXR confocal microscope at 10× magnification. A trained examiner graded each histological section in a blinded manner with excellent inter-rater reliability (intraclass correlation coefficient (2,1), 0.885; 95% confidence interval, 0.814–0.925) with another trained examiner.

Microarray analysis to assess transcriptomic response in murine chondrocytes to soft and stiff substrate

Archived microarray data derived from primary chondrocytes isolated from neonatal Kunming mice (postnatal day 1) cultured on young-like soft (0.5–1 kPa; $n = 3$) or aged-like stiff (100 kPa; $n = 3$) pAAm substrates for 24 h were retrieved from the GEO database (GSE63615)[32]. Raw intensity data were background-corrected using the normexp method, $\log_2$-transformed and quantile-normalized using the limma package. Low-intensity probes were filtered before downstream analysis using an intensity-based filtering approach, as recommended in the limma user guide[69]. In brief, probes with low average expression across samples were removed, as genes with consistently low signal intensity are unlikely to be reliably detected as differentially expressed and typically lack sufficient biological or statistical support. This filtering step improves statistical power without loss of meaningful information[66]. Gene symbols were converted to uppercase, probes lacking valid gene annotations were excluded, and duplicate probes mapping to the same gene were collapsed by taking the median expression value. The resulting normalized and filtered expression matrix was used for differential expression analysis. Differential expression of genes between stiff and soft substrate conditions was assessed using linear models implemented in limma, followed by empirical Bayes moderation. Genes were considered differentially expressed on the basis of moderated statistics, and results were adjusted for multiple testing using the false-discovery rate (FDR).

To determine whether genes altered under aged-like stiff conditions were enriched for targets of exercise-sensitive EV miRNAs, gene set enrichment analysis (GSEA) was performed using gene rankings derived from limma moderated *t*-statistics. Gene sets were

defined as experimentally validated and predicted 43 target genes of exercise-responsive 16 miRNAs. Enrichment significance was assessed using permutation-based testing with FDR correction. Leading edge analysis was subsequently applied to significantly enriched miRNA target gene sets to identify core genes driving enrichment, thereby highlighting stiffness-regulated transcripts potentially regulated by exercise-sensitive miRNA networks.

Microarray analysis of miRNA expression in osteoarthritis

Archived miRNA microarray data derived from serum collected from patients with KOA ($n = 12$; mean age, 69.3 years) and age-matched healthy controls ($n = 12$; mean age, 64.3 years) were retrieved from the GEO database (GSE105027)[88]. Raw signal intensity files were downloaded and processed using a microarray-appropriate preprocessing pipeline implemented in limma. Processed signal values (gProcessedSignal) were extracted from each array file and merged into a single expression matrix. Probes with non-finite values across samples were removed prior to downstream analysis. To reduce noise and improve statistical power, low-intensity miRNAs were filtered using an intensity-based threshold, excluding probes with low mean signal across samples; $\log_2$ transformation was subsequently applied to stabilize variance. Duplicate probes mapping to the same miRNA were collapsed by taking the median expression value, a standard approach for microarray summarization. The resulting matrix was then subjected to quantile normalization using the normalizeBetweenArrays function in limma to ensure comparable signal distributions across samples. The normalized expression matrix was used for all downstream analyses.

For differential expression analysis, linear modeling was performed using limma, followed by empirical Bayes moderation of standard errors. Differentially expressed miRNAs were identified using moderated statistics, and multiple testing correction was applied using the FDR method. Visualization included volcano plots generated using the EnhancedVolcano package, displaying $\log_2$(fold change) versus raw P values with predefined thresholds for statistical significance and effect size. For focused analysis of the miRNA-29 family, normalized expression values were visualized using violin plots with overlaid individual data points to illustrate distribution and sample-level variability. Outliers were removed using an interquartile-range-based approach prior to visualization.

Unsupervised machine learning

Principal component analysis (PCA) was performed for data reduction and visualization of global phenotypic variation. The same PCA pipeline was applied across all analyses, with only the input variables differing between datasets. For morphological profiling, PCA was conducted on 53 quantified cellular shape descriptors. To identify variables contributing to each principal component, loading matrices representing correlations between original variables and PCs were extracted. PCA was used to visualize separation between (1) chondrocytes cultured on soft versus stiff pAAm substrates, and (2) chondrocytes cultured on stiff pAAm supplemented with EV^{pre} or EV^{post}. For molecular phenotyping, PCA was additionally performed using integrated fluorescence intensities of type II collagen, aggrecan, SOX9, MMP13, ADAMTS4, DNMT1 and α-Klotho. For all PCA analyses, features were z-score normalized prior to dimensionality reduction, and PCA was performed using the prcomp function in RStudio. The first two principal components were visualized to assess group-wise clustering and EV-mediated phenotypic rescue.

Global DNA methylation assay

Levels of 5 methylcytosine (5mC) in DNA were measured using the 5mC DNA ELISA Kit (Zymo Research). A total of 200 ng DNA was used and the percent of 5mC in the samples was quantified, normalized to total DNA and compared with the standard curve provided in the kit.

Real-time PCR

Five hundred nanograms of isolated RNA was reverse transcribed using iScript Advanced cDNA Synthesis Kit (Bio-Rad). The mRNA levels of the *KLOTHO* gene were quantified by SYBR Green-based real-time PCR (qPCR) using SsoAdvanced Universal SYBR Green Supermix (Bio-Rad). Klotho gene expression levels were normalized to the expression level of RPL19, and the fold change of Klotho relative to universal human reference RNA was calculated using $2^{-\Delta\Delta Ct}$ method. Primer sequences were: Klotho (NM_004795.4) forward 5′-CAAGTACTGGATCACCATCGACA-3′ and reverse 5′-AGACTTTGGCATGAGCCAGGAG-3′. RPL19 (NM_000981) forward 5′-AAGCCTGTGACGGTCCATTC-3′ and reverse 5′-CCC TTCCGCTTACCTATGCC-3′.

Methylation-specific PCR

Two hundred nanograms of genomic DNA was subjected to bisulfite conversion with an EZ DNA Methylation Kit (Zymo Research) followed by methylation-specific PCR using primers specific for methylated DNA of the *KLOTHO* promoter. Differentially methylated region of human *KLOTHO* promoter in 5-aza-dC-treated cells was reported[89]. Primer sequences were designed to amplify the region of −306 to −89 bp (NG_011485.1: 4685–4902). hMS-Kl-forward 5′-AAAGAGAATGAATTTGAGCGTTTAC-3′; hMS-Kl: reverse 5′-ACTC CGCTAACAATAATTACCTACG-3′. Fully methylated control DNA (Zymo Research) was used as a reference to calculate the percentage methylation of DNA samples.

Statistics and reproducibility

All statistical analyses were performed using R Studio (version 4.5.2). For data visualization, individual biological replications are shown as dots on violin plots to illustrate data distribution and variability. Normality was formally assessed using the Shapiro–Wilk test. For comparisons between two independent groups, two-tailed unpaired Student's t tests were used. For paired experimental designs, including within-animal comparisons of EV^{pre} versus EV^{post} in vivo and matched pre- and post-exercise EV conditions in vitro, two-tailed paired t-tests were applied. Linear regression analyses were conducted to assess relationships among in vitro chondrocyte outcomes, as well as associations between EV miRNA-29 levels and chondrocyte outcomes, including α-Klotho level. Model assumptions were evaluated by inspection of residuals versus fitted values and assessment of independence between observations. No correction for multiple comparisons was applied, as outcome measures were defined a priori and were highly correlated. In all experiments, $P < 0.05$ was considered statistically significant. Unless otherwise indicated, n denotes the number of independent biological observations. Additional details regarding data representation and statistical procedures are provided in the figure legends.

Although there are no established guidelines for in vitro studies, we have adapted the essential 10 from the Animal Research Reporting In Vivo Experiments (ARRIVE) framework. We excluded randomization (item 4) and experimental animals (item 8) when they were not applicable[90]. This study also followed the reporting guideline proposed by Emmerich et al[91]. Where appropriate, samples or animals were randomized or assigned to experimental conditions to minimize potential bias using a computer-generated permuted block randomization scheme. Experimental conditions, sample processing and data collection were standardized across groups, and experimental order or environmental factors were randomized where feasible using a computer-generated randomization scheme. Where possible, a power analysis based on pilot data was done to select the number of animals needed for the study using Power and Sample Size Program (version 3.1.2; Vanderbilt University Medical Center)[92]. For example, sample-size calculations estimated that 10 older adults undergoing aerobic exercise were required to achieve a statistical power of 0.8 based on type II collagen expression in vitro and histologically defined cartilage integrity in vivo. For experiments for which a priori power analysis

was not performed, no statistical methods were used to predetermine sample sizes, and sample sizes were selected on the basis of those used in previous publications. The treatments and image analyses of EVs isolated before and after exercise were conducted under the same conditions. For in vivo experiments, cage locations were changed weekly to minimize potential environmental bias. All histological outcome assessments were conducted in a blinded manner. We have submitted all relevant data of EV experiments to the EV-TRACK knowledgebase (EV-TRACK ID: EV260074).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All omics datasets analyzed in this study were obtained from publicly available repositories and are described in the paper. Any additional information required to reanalyze the data presented in this work is available from the corresponding authors upon reasonable request. Source data are provided with this paper.

References

1. Denham, J. & Spencer, S. J. Emerging roles of extracellular vesicles in the intercellular communication for exercise-induced adaptations. *Am. J. Physiol. Endocrinol. Metab.* **319**, E320–E329 (2020).
2. Zaborowski, M. P., Balaj, L., Breakefield, X. O. & Lai, C. P. Extracellular vesicles: composition, biological relevance, and methods of study. *Bioscience* **65**, 783–797 (2015).
3. Nair, V. D. et al. Sedentary and trained older men have distinct circulating exosomal microRNA profiles at baseline and in response to acute exercise. *Front. Physiol.* **11**, 605 (2020).
4. Vanderboom, P. M. et al. A size-exclusion-based approach for purifying extracellular vesicles from human plasma. *Cell Rep. Methods* **1**, 100055 (2021).
5. Bei, Y. et al. Exercise-induced circulating extracellular vesicles protect against cardiac ischemia-reperfusion injury. *Basic Res. Cardiol.* **112**, 38 (2017).
6. Safdar, A., Saleem, A. & Tarnopolsky, M. A. The potential of endurance exercise-derived exosomes to treat metabolic diseases. *Nat. Rev. Endocrinol.* **12**, 504–517 (2016).
7. Pegtel, D. M. et al. Functional delivery of viral miRNAs via exosomes. *Proc. Natl Acad. Sci. USA* **107**, 6328–6333 (2010).
8. Herranz, H. & Cohen, S. M. MicroRNAs and gene regulatory networks: managing the impact of noise in biological systems. *Genes Dev.* **24**, 1339–1344 (2010).
9. Arner, P. & Kulyté, A. MicroRNA regulatory networks in human adipose tissue and obesity. *Nat. Rev. Endocrinol.* **11**, 276–288 (2015).
10. Wong, A. K., Krishnan, A. & Troyanskaya, O. G. GIANT 2.0: genome-scale integrated analysis of gene networks in tissues. *Nucleic Acids Res.* **46**, W65–W70 (2018).
11. Lage, K. et al. A large-scale analysis of tissue-specific pathology and gene expression of human disease genes and complexes. *Proc. Natl Acad. Sci. USA* **105**, 20870–20875 (2008).
12. Loeser, R. F., Collins, J. A. & Diekman, B. O. Ageing and the pathogenesis of osteoarthritis. *Nat. Rev. Rheumatol.* **12**, 412–420 (2016).
13. GBD 2021 Osteoarthritis Collaborators. Global, regional, and national burden of osteoarthritis, 1990–2020 and projections to 2050: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet Rheumatol.* **5**, e508–e522 (2023).
14. McAlindon, T. E. et al. OARSI guidelines for the non-surgical management of knee osteoarthritis. *Osteoarthr. Cartil.* **22**, 363–388 (2014).
15. Iijima, H. et al. Physiological exercise loading suppresses post-traumatic osteoarthritis progression via an increase in bone morphogenetic proteins expression in an experimental rat knee model. *Osteoarthr. Cartil.* **6**, 964–975 (2017).
16. Rando, T. A. & Ambrosio, F. Regenerative rehabilitation: applied biophysics meets stem cell therapeutics. *Cell Stem Cell* **22**, 306–309 (2018).
17. Kern, F. et al. miEAA 2.0: integrating multi-species microRNA enrichment analysis and workflow management systems. *Nucleic Acids Res.* **48**, W521–W528 (2020).
18. Huang, H.-Y. et al. miRTarBase 2020: updates to the experimentally validated microRNA–target interaction database. *Nucleic Acids Res.* **48**, D148–D154 (2020).
19. Berrier, A. L. & Yamada, K. M. Cell–matrix adhesion. *J. Cell Physiol.* **213**, 565–573 (2007).
20. Kriegel, A. J., Liu, Y., Fang, Y., Ding, X. & Liang, M. The miR-29 family: genomics, cell biology, and relevance to renal and cardiovascular injury. *Physiol. Genomics* **44**, 237–244 (2012).
21. Sealfon, R. S. G. et al. HumanBase: an interactive AI platform for human biology. *Nat. Methods* **23**, 483–484 (2026).
22. Zhao, Z. et al. Mechanotransduction pathways in the regulation of cartilage chondrocyte homoeostasis. *J. Cell. Mol. Med.* **24**, 5408–5419 (2020).
23. Knight, A. & Levick, J. R. Morphometry of the ultrastructure of the blood–joint barrier in the rabbit knee. *Q. J. Exp. Physiol.* **69**, 271–288 (1984).
24. Valdeolivas, A. et al. Random walk with restart on multiplex and heterogeneous biological networks. *Bioinformatics* **35**, 497–505 (2019).
25. Iijima, H. et al. Age-related matrix stiffening epigenetically regulates α-Klotho expression and compromises chondrocyte integrity. *Nat. Commun.* **14**, 18 (2023).
26. Phillip, J. M., Aifuwa, I., Walston, J. & Wirtz, D. The mechanobiology of aging. *Annu. Rev. Biomed. Eng.* **17**, 113–141 (2015).
27. Schmauck-Medina, T. et al. New hallmarks of ageing: a 2022 Copenhagen ageing meeting summary. *Aging* **14**, 6829–6839 (2022).
28. Stolz, M. et al. Early detection of aging cartilage and osteoarthritis in mice and patient samples using atomic force microscopy. *Nat. Nanotechnol.* **4**, 186–192 (2009).
29. Selman, M. & Pardo, A. Fibroageing: an ageing pathological feature driven by dysregulated extracellular matrix-cell mechanobiology. *Ageing Res. Rev.* **70**, 101393 (2021).
30. Kim, J. H. et al. Matrix cross-linking-mediated mechanotransduction promotes posttraumatic osteoarthritis. *Proc. Natl Acad. Sci. USA* **112**, 9424–9429 (2015).
31. Zhong, W. et al. YAP-mediated regulation of the chondrogenic phenotype in response to matrix elasticity. *J. Mol. Histol.* **44**, 587–595 (2013).
32. Du, J. et al. Extracellular matrix stiffness dictates Wnt expression through integrin pathway. *Sci. Rep.* **6**, 20395 (2016).
33. Zhou, C. et al. Microenvironmental stiffness mediates cytoskeleton re-organization in chondrocytes through laminin-FAK mechanotransduction. *Int J. Oral Sci.* **14**, 15 (2022).
34. Swift, D. L., Johannsen, N. M., Lavie, C. J., Earnest, C. P. & Church, T. S. The role of exercise and physical activity in weight loss and maintenance. *Prog. Cardiovasc. Dis.* **56**, 441–447 (2014).
35. Sahu, A. et al. Regulation of aged skeletal muscle regeneration by circulating extracellular vesicles. *Nat. Aging* **1**, 1148–1161 (2021).
36. Welsh, J. A. et al. Minimal information for studies of extracellular vesicles (MISEV2023): from basic to advanced approaches. *J. Extracell. Vesicles* **13**, e12404 (2024).
37. Estébanez, B., Jiménez-Pavón, D., Huang, C. J., Cuevas, M. J. & González-Gallego, J. Effects of exercise on exosome release and cargo in in vivo and ex vivo models: a systematic review. *J. Cell Physiol.* **236**, 3336–3353 (2021).

38. Carpenter, A. E. et al. CellProfiler: image analysis software for identifying and quantifying cell phenotypes. *Genome Biol.* **7**, R100 (2006).
39. Loeser, R. F. Integrins and chondrocyte-matrix interactions in articular cartilage. *Matrix Biol.* **39**, 11–16 (2014).
40. Hynes, R. O. Integrins: bidirectional, allosteric signaling machines. *Cell* **110**, 673–687 (2002).
41. Kanamoto, T. et al. Integrin α2β1 plays an important role in the interaction between human articular cartilage-derived chondrocytes and atelocollagen gel. *Sci. Rep.* **11**, 1757 (2021).
42. Kanehisa, M., Sato, Y., Kawashima, M., Furumichi, M. & Tanabe, M. KEGG as a reference resource for gene and protein annotation. *Nucleic Acids Res.* **44**, D457–D462 (2016).
43. Moore, L. D., Le, T. & Fan, G. DNA methylation and its basic function. *Neuropsychopharmacology* **38**, 23–38 (2013).
44. Iijima, H. et al. Meta-analysis integrated with multi-omics data analysis to elucidate pathogenic mechanisms of age-related knee osteoarthritis in mice. *J. Gerontol. A Biol. Sci. Med. Sci.* **77**, 1321–1334 (2022).
45. Jackson, S. J. et al. Does age matter? The impact of rodent age on study outcomes. *Lab. Anim.* **51**, 160–169 (2017).
46. Chowdhary, K. et al. Aging affects the efficacy of platelet-rich plasma treatment for osteoarthritis. *Am. J. Phys. Med. Rehabil.* **102**, 597–604 (2023).
47. Liu, Y. et al. Renal medullary microRNAs in Dahl salt-sensitive rats: miR-29b regulates several collagens and related genes. *Hypertension* **55**, 974–982 (2010).
48. Ramdas, V., McBride, M., Denby, L. & Baker, A. H. Canonical transforming growth factor-β signaling regulates disintegrin metalloprotease expression in experimental renal fibrosis via miR-29. *Am. J. Pathol.* **183**, 1885–1896 (2013).
49. Cheng, J., Wang, Y., Wang, D. & Wu, Y. Identification of collagen 1 as a post-transcriptional target of miR-29b in skin fibroblasts: therapeutic implication for scar reduction. *Am. J. Med. Sci.* **346**, 98–103 (2013).
50. Montgomery, R. L. et al. MicroRNA mimicry blocks pulmonary fibrosis. *EMBO Mol. Med.* **6**, 1347–1356 (2014).
51. Ko, J.-Y. et al. MicroRNA-29a counteracts synovitis in knee osteoarthritis pathogenesis by targeting VEGF. *Sci. Rep.* **7**, 3584 (2017).
52. Kawanishi, N. et al. Exercise training attenuates hepatic inflammation, fibrosis and macrophage infiltration during diet-induced obesity in mice. *Brain Behav. Immun.* **26**, 931–941 (2012).
53. Li, X., Zhen, Z., Tang, G., Zheng, C. & Yang, G. MiR-29a and MiR-140 protect chondrocytes against the anti-proliferation and cell matrix signaling changes by IL-1β. *Mol. Cells* **39**, 103–110 (2016).
54. Wang, H. et al. Exosome-mediated miR-29 transfer reduces muscle atrophy and kidney fibrosis in mice. *Mol. Ther.* **27**, 571–583 (2019).
55. Lin, E. Y. et al. Engineered exosomes containing microRNA-29b-2 and targeting the somatostatin receptor reduce presenilin 1 expression and decrease the β-amyloid accumulation in the brains of mice with Alzheimer's disease. *Int. J. Nanomed.* **19**, 4977–4994 (2024).
56. Chen, R. et al. Delivery of engineered extracellular vesicles with miR-29b editing system for muscle atrophy therapy. *J. Nanobiotechnology* **20**, 304 (2022).
57. Lampi, M. C. & Reinhart-King, C. A. Targeting extracellular matrix stiffness to attenuate disease: from molecular mechanisms to clinical trials. *Sci. Transl. Med.* **10**, eaao0475 (2018).
58. Guo, S. et al. Stimulating extracellular vesicles production from engineered tissues by mechanical forces. *Nano Lett.* **21**, 2497–2504 (2021).
59. Najrana, T. et al. Mechanical stretch regulates the expression of specific miRNA in extracellular vesicles released from lung epithelial cells. *J. Cell Physiol.* **235**, 8210–8223 (2020).
60. Yamagata, M., Kimura, T., Chang, A. H. & Iijima, H. Sex differences in ambulatory biomechanics: a meta-analysis providing a mechanistic insight into knee osteoarthritis. *Med. Sci. Sports Exerc.* **57**, 144–153 (2025).
61. Vechetti, I. J. Jr, Valentino, T., Mobley, C. B. & McCarthy, J. J. The role of extracellular vesicles in skeletal muscle and systematic adaptation to exercise. *J. Physiol.* **599**, 845–861 (2021).
62. Safdar, A. & Tarnopolsky, M. A. Exosomes as mediators of the systemic adaptations to endurance exercise. *Cold Spring Harb. Perspect. Med.* **8**, a029827 (2018).
63. Kotewitsch, M., Heimer, M., Schmitz, B. & Mooren, F. C. Non-coding RNAs in exercise immunology: a systematic review. *J. Sport Health Sci.* **13**, 311–338 (2024).
64. Sun, L. et al. GEPREP: A comprehensive data atlas of RNA-seq-based gene expression profiles of exercise responses. *J. Sport Health Sci.* **14**, 100992 (2025).
65. Law, C. W. et al. RNA-seq analysis is easy as 1-2-3 with limma, Glimma and edgeR. *F1000Res* **5**, ISCB Comm J-1408 (2016).
66. Chen, Y., Lun, A. T. & Smyth, G. K. From reads to genes to pathways: differential expression analysis of RNA-Seq experiments using Rsubread and the edgeR quasi-likelihood pipeline. *F1000Res* **5**, 1438 (2016).
67. Robinson, M. D., McCarthy, D. J. & Smyth, G. K. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* **26**, 139–140 (2010).
68. Law, C. W., Chen, Y., Shi, W. & Smyth, G. K. voom: precision weights unlock linear model analysis tools for RNA-seq read counts. *Genome Biol.* **15**, R29 (2014).
69. Ritchie, M. E. et al. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* **43**, e47 (2015).
70. Iijima, H., Zhang, F., Ambrosio, F. & Matsui, Y. Network-based cytokine inference implicates Oncostatin M as a driver of an inflammation phenotype in knee osteoarthritis. *Aging Cell* **23**, e14043 (2024).
71. Köhler, S., Bauer, S., Horn, D. & Robinson, P. N. Walking the interactome for prioritization of candidate disease genes. *Am. J. Hum. Genet.* **82**, 949–958 (2008).
72. Iijima, H., Ambrosio, F. & Matsui, Y. Network-based systematic dissection of exercise-induced inhibition of myosteatosis in older individuals. *J. Physiol.* **603**, 45–67 (2025).
73. McNeill, T. J., Ambrosio, F. & Iijima, H. Network model to predict age-related transcriptional reprogramming. *Aging Cell* **25**, e70683 (2026).
74. Katz, N. B. et al. Relaxin-2 as a modifiable hormonal pathway in knee osteoarthritis: dose dependent effects on chondrocyte phenotype. *PM R* **18**, S92–S103 (2026).
75. Yu, G., Wang, L. G., Han, Y. & He, Q. Y. clusterProfiler: an R package for comparing biological themes among gene clusters. *Omics* **16**, 284–287 (2012).
76. Moore, H. M. et al. Biospecimen reporting for improved study quality (BRISQ). *J. Proteome Res.* **10**, 3429–3438 (2011).
77. Lucien, F. et al. MIBlood-EV: minimal information to enhance the quality and reproducibility of blood extracellular vesicle research. *J. Extracell. Vesicles* **12**, e12385 (2023).
78. Théry, C. et al. Minimal information for studies of extracellular vesicles 2018 (MISEV2018): a position statement of the International Society for Extracellular Vesicles and update of the MISEV2014 guidelines. *J. Extracell. Vesicles* **7**, 1535750 (2018).
79. Wang, K. et al. Substrate stiffness-dependent carbon nanotube-induced lung fibrogenesis. *Nano Lett.* **19**, 5443–5451 (2019).
80. Sahu, A. et al. Age-related declines in α-Klotho drive progenitor cell mitochondrial dysfunction and impaired muscle regeneration. *Nat. Commun.* **9**, 4859 (2018).

81. Bolte, S. & Cordelières, F. P. A guided tour into subcellular colocalization analysis in light microscopy. *J. Microsc.* **224**, 213–232 (2006).
82. Dunn, K. W., Kamocka, M. M. & McDonald, J. H. A practical guide to evaluating colocalization in biological microscopy. *Am. J. Physiol. Cell Physiol.* **300**, C723–C742 (2011).
83. Flurkey, K., Currer, J. M. & Harrison, D. Mouse models in aging research. in *The Mouse in Biomedical Research* 637–672 (Elsevier, 2007).
84. Glasson, S. S., Chambers, M. G., Van Den Berg, W. B. & Little, C. B. The OARSI histopathology initiative—recommendations for histological assessments of osteoarthritis in the mouse. *Osteoarthr. Cartil.* **18**, S17–S23 (2010).
85. Hayashi, D. et al. Pre-radiographic osteoarthritic changes are highly prevalent in the medial patella and medial posterior femur in older persons: Framingham OA study. *Osteoarthr. Cartil.* **22**, 76–83 (2014).
86. Armstrong, A. R., Carlson, C. S., Rendahl, A. K. & Loeser, R. F. Optimization of histologic grading schemes in spontaneous and surgically-induced murine models of osteoarthritis. *Osteoarthr. Cartil.* **29**, 536–546 (2021).
87. Pritzker, K. P. et al. Osteoarthritis cartilage histopathology: grading and staging. *Osteoarthr. Cartil.* **14**, 13–29 (2006).
88. Ntoumou, E. et al. Serum microRNA array analysis identifies miR-140-3p, miR-33b-3p and miR-671-3p as potential osteoarthritis biomarkers involved in metabolic processes. *Clin. Epigenetics* **9**, 127 (2017).
89. Sun, Y.-C., Chang, S.-C. & Wu, M.-S. Suppression of Klotho expression by protein-bound uremic toxins is associated with increased DNA methyltransferase expression and DNA hypermethylation. *Kidney Int.* **81**, 640–650 (2012).
90. Percie du Sert, N. et al. The ARRIVE guidelines 2.0: updated guidelines for reporting animal research. *PLoS Biol.* **18**, e3000410 (2020).
91. Emmerich, C. H. & Harris, C. M. Minimum information and quality standards for conducting, reporting, and organizing in vitro research. *Handb. Exp. Pharmacol.* **257**, 177–196 (2020).
92. Dupont, W. D. & Plummer, W. D. Jr. Power and sample size calculations for studies involving linear regression. *Control Clin. Trials* **19**, 589–601 (1998).

Acknowledgements

We thank the study participants and the clinical research team at the University of Pittsburgh for their contributions to thisstudy. We also thank the laboratory members at the Discovery Center for Musculoskeletal Recovery at Spaulding for theirvaluable advice and assistance.

Author contributions

All authors made substantial contributions in the following areas: (1) conception and design of the study, acquisition of data, analysis and interpretation of data and drafting of the article; (2) final approval of the article version to be submitted; and (3) agreement to be personally accountable for the author's own contributions and to ensure that questions related to the accuracy are appropriately investigated, resolved, and the resolution documented in the literature. The specific contributions of the authors are as follows: H.I., K.M.D. and F.A. provided the concept, idea and experimental design for the studies. H.I. and F.A. wrote the paper. H.I., K.M.D., K.W., A.M.K., E.A., W.-Y.T, R.J.R., J.M.J and F.A. contributed to data collection, analyses, interpretation and review of the paper. H.I. and F.A. obtained funding for the studies.

Funding

This study was supported in part by a research grant from the Uehara Memorial Foundation to H.I.; a grant from the Tokai Pathways to Global Excellence Project (T-GEx) (https://www.t-gex.nagoya-u.ac.jp/en/) to H.I.; a JSPS KAKENHI (grant. no. 23H03308) from the Japan Society for the Promotion of Science (https://www.jsps.go.jp/) to H.I.; NIA R01AG052978, NIA R01AG061005 and R01AG087565 to F.A.; NIA R01AG089455 and NIA R01AG092299 to H.I. and F.A.; and NIA R01AG090356 to H.I. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s43587-026-01225-9.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43587-026-01225-9.

Correspondence and requests for materials should be addressed to Hirotaka Iijima or Fabrisia Ambrosio.

Peer review information *Nature Aging* thanks Mandy Peffers, who co-reviewed with Emily Clarke; Fernanda De Felice; and Ge Zhang for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

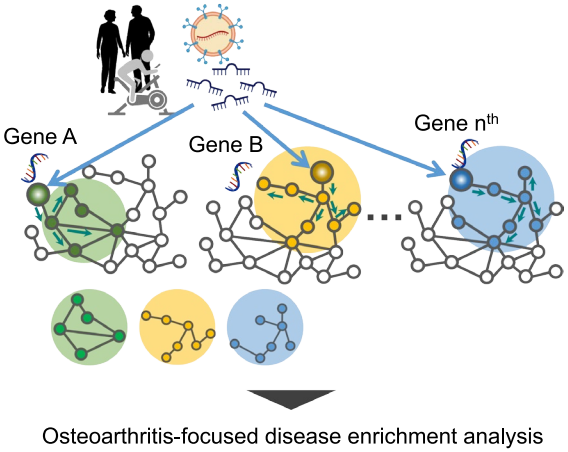
Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

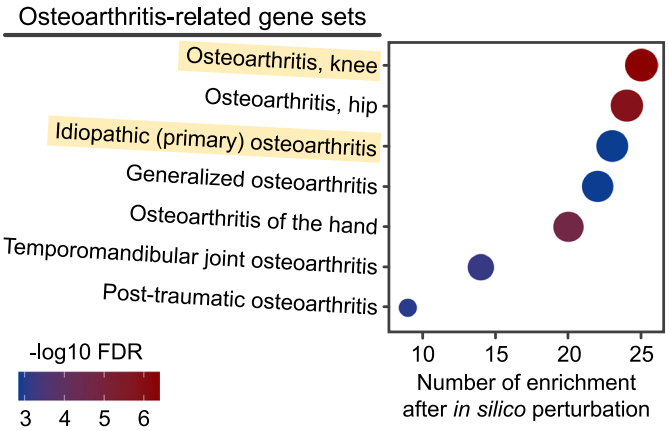
Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2026

https://doi.org/10.1038/s43587-026-01225-9

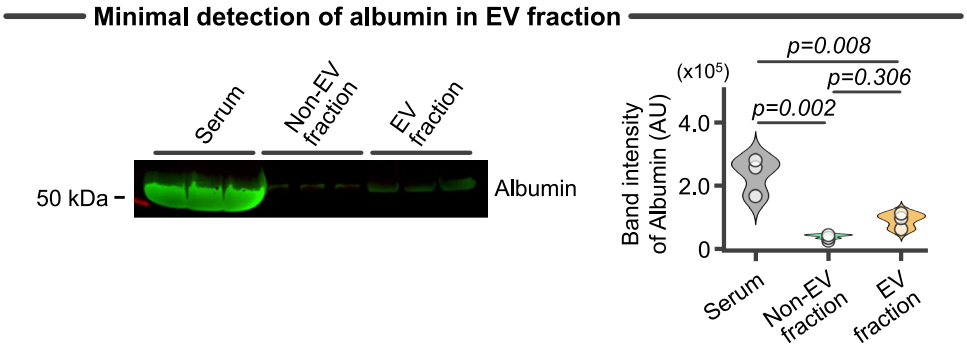
Network propagation clarified EV miRNA-induced secondary impact on osteoarthritis of knee joint





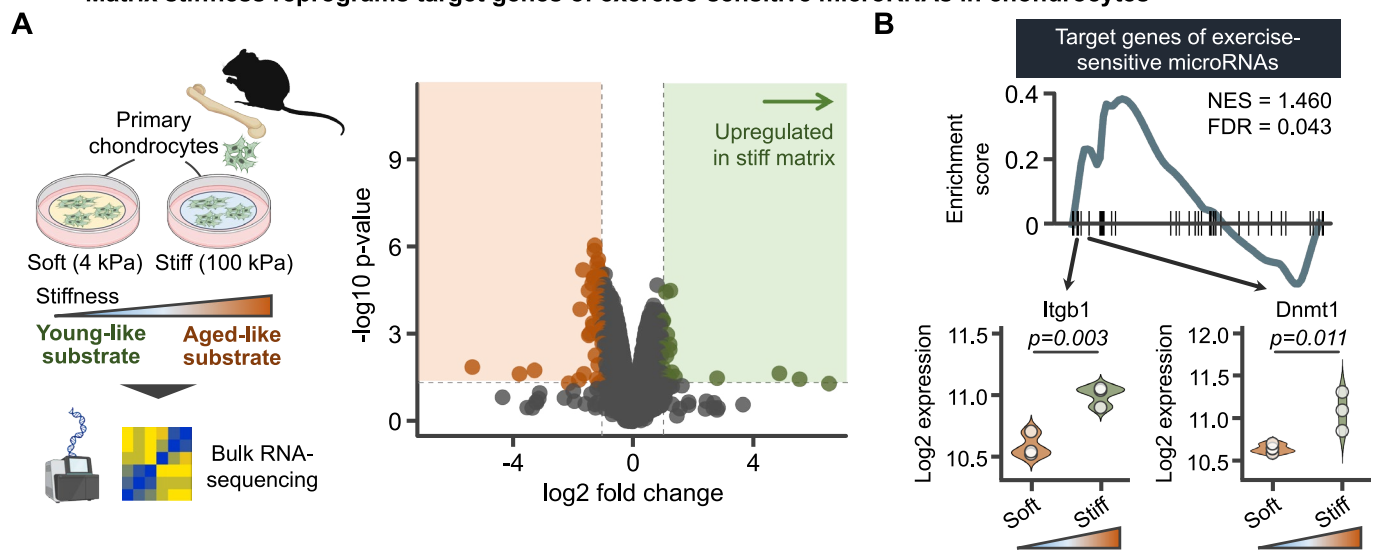
Extended Data Fig. 1 | Disease-enrichment analysis of pseudo-activated genes following exercise-primed EV miRNA perturbation. In silico perturbation of the cartilage-specific network revealed that exercise-primed EV miRNAs preferentially influence genes associated with knee osteoarthritis and idiopathic (primary) osteoarthritis, whereas post-traumatic osteoarthritis exhibited comparatively lower enrichment. The dot size represents the absolute impact score, and the color intensity indicates the average −log10 (FDR). This analysis highlights primary osteoarthritis, specifically age-related knee osteoarthritis, as the relevant clinical model in this study. Data visualization was performed using BioRender Graphing (BioRender, Toronto, ON, Canada; https://www.biorender.com, accessed March 2026; RRID:SCR_018361). All raw data supporting these findings are available in the Supplementary Data. Abbreviation: EVs, extracellular vesicles; FDR, false discovery rate; miRNA, microRNA.

https://doi.org/10.1038/s43587-026-01225-9



Extended Data Fig. 2 | Detection of albumin protein in EV fractions. Albumin, the most abundant plasma protein, was detected at low levels in non-EV and EV fractions compared with serum (n = 3 per group). The very low levels of albumin in EV fractions suggest that they are largely free of major plasma protein contamination and that downstream analyses are unlikely to be confounded by residual albumin. Samples shown in each blot were derived from the same experiment and processed in parallel. Statistical analysis was performed using analysis of variance with post-hoc Tukey–Kramer test. All raw data, including unadjusted and intensity-enhanced uncropped Western blot images with clearly visible molecular weight markers, are available in the Supplementary Data. Abbreviations: EV, extracellular vesicles.

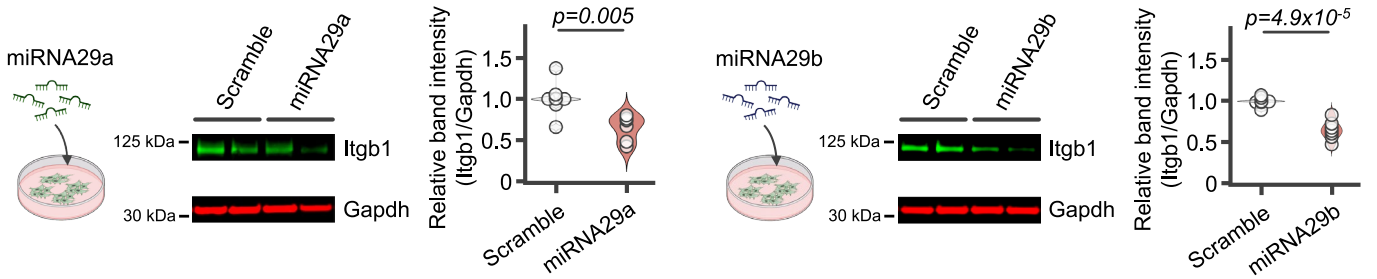
https://doi.org/10.1038/s43587-026-01225-9



Extended Data Fig. 3 | Stiff substrate upregulates target genes of exercise-sensitive microRNAs, including *Itgb1* and *Dnmt1*. **a**, Schematic of the experimental design and volcano plot of differential gene expression analysis using the publicly available RNA-sequencing dataset (GSE63615), comparing primary chondrocytes cultured on soft (4 kPa; n = 3) versus stiff (100 kPa; n = 3) substrates. **b**, GSEA demonstrating significant enrichment of 43 target genes of 16 exercise-sensitive microRNAs in chondrocytes cultured on stiff substrate. Itgb1 and Dnmt1 were identified among the leading-edge genes. Violin plots show increased expression of Itgb1 and Dnmt1 under stiff substrates compared with soft counterparts. Statistical analysis was performed using the R package limma based on a two-sided empirical Bayes-moderated t-test (**a**) and a two-tailed Student t-test (**b**). Data visualization was performed using BioRender Graphing (BioRender, Toronto, ON, Canada; https://www.biorender.com, accessed March 2026; RRID:SCR_018361). All raw data supporting these findings are available in the Supplementary Data. Abbreviations: GSEA, gene set enrichment analysis; NES, normalized enrichment score; FDR, false discovery rate.

https://doi.org/10.1038/s43587-026-01225-9

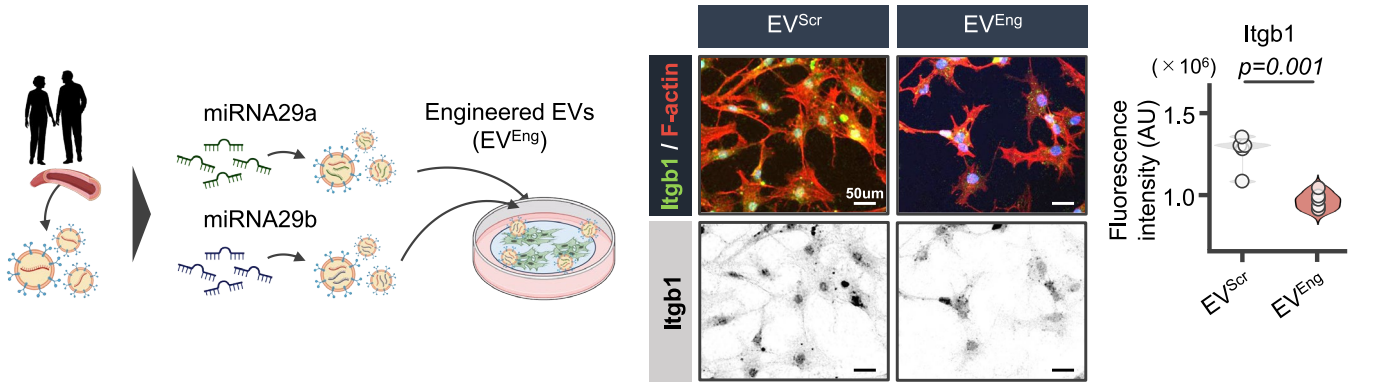
Transfection of miRNA29 family inhibited target genes in aged human chondrocytes



Extended Data Fig. 4 | Transfection of miRNA-29a-3p and miRNA-29b-3p reduced Itgb1 protein level. Transfection of miRNA-29a-3p (scramble, n = 7; miRNA-29a, n = 6) and miRNA-29b-3p (n = 7 per group) mimics into aged human chondrocytes significantly reduced Itgb1 protein expression, as demonstrated by western blot. Samples shown in each blot were derived from the same experiment and processed in parallel. Statistical analyses were performed using two-tailed Student's t-tests. Data visualization was performed using BioRender Graphing (BioRender, Toronto, ON, Canada; https://www.biorender.com, accessed March 2026; RRID:SCR_018361). All raw data, including unadjusted and intensity-enhanced uncropped Western blot images with clearly visible molecular weight markers, are available in the Supplementary Data. Abbreviations: miRNA, microRNA.

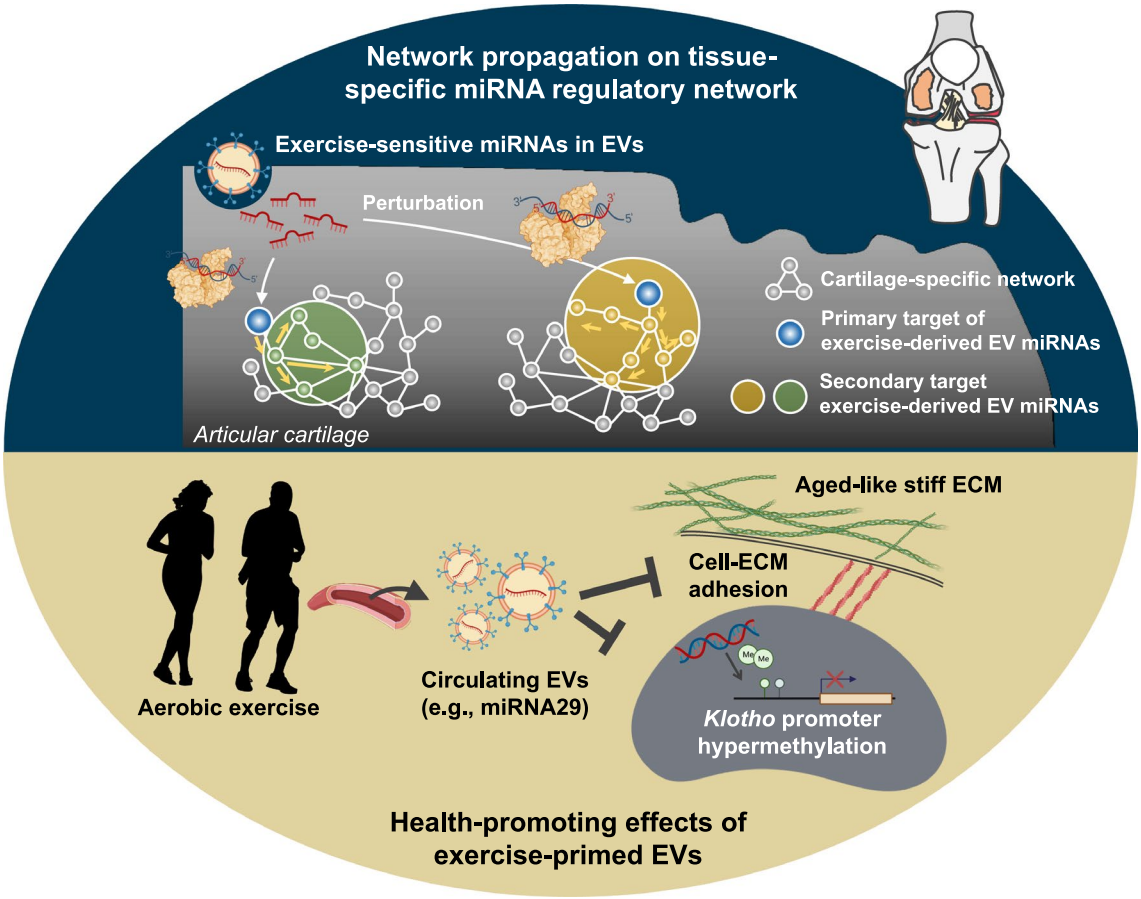
https://doi.org/10.1038/s43587-026-01225-9

Engineered EVs counteracted stiff matrix-induced increase in Itgb1 levels of chondrocytes



Extended Data Fig. 5 | Engineered EVs suppressed stiff matrix–induced Itgb1 expression in chondrocytes. Treatment with engineered miRNA-29-enriched EVs significantly reduced expression in chondrocytes compared to scramble counterparts (n = 5 per group; 1,000–1,500 cells analyzed per individual sample). Scale bar: 50µm. Statistical analysis was performed using a two-tailed Student t-test. Data visualization was performed using BioRender Graphing (BioRender, Toronto, ON, Canada; https://www.biorender.com, accessed March 2026; RRID:SCR_018361). All raw data supporting these findings are available in the Supplementary Data. Abbreviations: EV, extracellular vesicle; EV^{Eng}, engineered EV; EV^{scr}, scramble EV; miRNA, microRNA.

https://doi.org/10.1038/s43587-026-01225-9



Extended Data Fig. 6 | Graphical abstract. This study introduced a network paradigm to simulate EV-mediated miRNA regulatory network perturbation in recipient articular chondrocyte and uncovered cellular phenotypic shifts induced by exercise-primed EVs. Network perturbation analysis highlighted the miRNA-29 family as a dominant regulator of these effects. The health-promoting effects of exercise-primed EVs were attributed, at least in part, to epigenetic reprogramming of the gene encoding the longevity factor α-Klotho, establishing a mechanistic link between aerobic exercise, EV miRNA-29 cargo, and modulation of cartilage aging. Data visualization was performed using BioRender Graphing (BioRender, Toronto, ON, Canada; https://www.biorender.com, accessed March 2026; RRID:SCR_018361).

nature portfolio

Corresponding author(s): Hirotaka Iijima, Fabrisia Ambrosio

Last updated by author(s): Aug 18, 2026

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	EV size distribution and particle concentration were measured using nanoparticle tracking analysis (NTA) on a NanoSight LM10 system with NanoSight NTA software (version 3.3). Digital PCR data for miRNA-29a-3p and miRNA-29b-3p were acquired and analyzed using QIAcuity software (QIAGEN). Fluorescence images were acquired using a Nikon AXR confocal microscope, and quantitative image analyses were performed using ImageJ/FIJI and CellProfiler (version 4.2.6).
Data analysis	Statistical analyses were performed using R (version 4.5.2). Differential expression analyses of publicly available microarray datasets were performed using the limma package, with false discovery rate (FDR) correction for multiple testing. Gene set enrichment analysis was performed using established GSEA methods. Principal component analysis (PCA) was performed using the prcomp function in R. Quantitative image analyses were performed using ImageJ/FIJI, including Mander's overlap coefficient for colocalization analyses, and CellProfiler for cellular morphology profiling. No new computational method or software was developed in this study; the analyses were performed by combining established and publicly available analytical methods and software packages. All omics datasets analyzed in this study were obtained from publicly available repositories, and the corresponding accession numbers are provided in the manuscript.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All omics datasets analyzed in this study were obtained from publicly available repositories and are described in the manuscript. The source data underlying the figures and the findings reported in this study are provided with the paper. Any additional information required to reanalyze the data presented in this work is available from the corresponding author upon reasonable request.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	The aerobic exercise cohort included 10 participants (80% women). Sex was self-reported at study enrollment. Gender identity was not separately assessed.
Reporting on race, ethnicity, or other socially relevant groupings	Race and ethnicity were not used as eligibility criteria or for participant selection in this study.
Population characteristics	The cohort comprised community-dwelling, sedentary older adults aged 65–80 years (n = 10; mean age, 69.4 ± 3.7 years; 80% women). Participants were independently ambulatory and had physician clearance for participation.
Recruitment	Participants were recruited from the community, primarily through a clinical registry of individuals interested in research participation. Initial eligibility was assessed by telephone screening, followed by an in-person study orientation and written informed consent. Participants were required to be sedentary, aged 65–80 years, independently ambulatory, and able to provide informed consent.
Ethics oversight	The study protocol was reviewed and approved by the Institutional Review Board of the University of Pittsburgh. All participants provided written informed consent before enrollment and study procedures. Participants did not receive financial or other compensation. The cohort was derived from the Cognition and Exercise (CogEx) trial (ClinicalTrials.gov identifier: NCT03696082).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes were determined using pilot data and prior literature when possible. For EV analyses related to exercise responses, power analysis indicated that samples from 10 participants were sufficient to achieve 80% power.
Data exclusions	Samples or animals were excluded only when predefined quality or health criteria were met, including visible hemolysis or serious health abnormalities. One mouse was excluded due to severe knee joint inflammation and cartilage loss after intra-articular injection.
Replication	Experiments were performed using independent biological replicates, with experimental conditions and sample processing standardized across groups.
Randomization	Where appropriate, samples or animals were randomized to experimental conditions using computer-generated randomization schemes, with or without permuted blocks, as applicable. Treatment assignment was not randomized in the intra-articular EV experiment because EVpost and EVpre were consistently administered to the right and left knees, respectively, using a within-animal design.
Blinding	Histological assessments and imaging analyses were conducted by investigators blinded to group allocation.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Primary antibodies used in this study included antibodies against type II collagen, aggrecan, Sox9, Mmp13, Adamts4, Itgb1, Dnmt1, and α-Klotho for immunofluorescence and protein detection assays. Detailed information regarding antibody sources, catalog numbers, and dilution conditions is provided in the Methods section.
Validation	All antibodies were used according to the manufacturers' recommendations. Specificity of immunostaining was confirmed by including negative control samples in which the primary antibody was omitted.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Young (4–6 months) and aged (18–24 months) male C57BL/6N mice obtained from the NIA Rodent Colony were used.
Wild animals	No wild animals were used.
Reporting on sex	Male mice were used throughout the animal experiments.
Field-collected samples	No field-collected samples were used.
Ethics oversight	All animal experiments were approved by the Massachusetts General Hospital Institutional Animal Care and Use Committees (Protocol IDs: 2022N000171 and 2022N000148).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	The human aerobic exercise cohort was derived from the Cognition and Exercise (CogEx) trial, a randomized controlled trial registered at ClinicalTrials.gov (NCT03696082). The present EV analyses were exploratory analyses of biospecimens collected during the registered trial and were not prespecified clinical trial outcomes.
Study protocol	The CogEx study protocol was reviewed and approved by the Institutional Review Board of the University of Pittsburgh. All participants provided written informed consent before enrollment and study procedures. The trial was designed to investigate the effects of exercise interventions in community-dwelling older adults. The present study used serum samples collected according to the original CogEx study protocol.
Data collection	Participants were community-dwelling adults aged 65–80 years who were sedentary at baseline and met predefined eligibility criteria. Participants underwent a 12-week aerobic exercise intervention consisting of one supervised and four home-based sessions per week. Serum samples were collected in the morning following a 10–12 h fast and avoidance of strenuous physical activity for 10–12 h before collection. Blood was processed according to a standardized protocol and serum samples were stored at −80°C until EV isolation. The aerobic exercise group included 10 participants (mean age, 69.4 ± 3.7 years; 80% female). Detailed participant characteristics are provided in Table S2.
Outcomes	The primary outcomes of the registered CogEx trial were not the focus of the present study. The analyses reported here were

Outcomes	exploratory investigations of exercise-associated changes in circulating extracellular vesicles and their effects on chondrocyte phenotypes. EV analyses included EV characterization, miRNA-29 quantification, and assessment of the effects of pre- and post-exercise EVs on human chondrocyte phenotypes. The 12-week exercise intervention and biospecimen collection were completed as planned. The CogEx trial was subsequently terminated because institutional funding support was discontinued; this was unrelated to adverse events or participant safety. Consequently, planned long-term follow-up outcomes were not collected and are not reported in this manuscript.

Plants

Seed stocks	Not applicable
Novel plant genotypes	Not applicable
Authentication	Not applicable